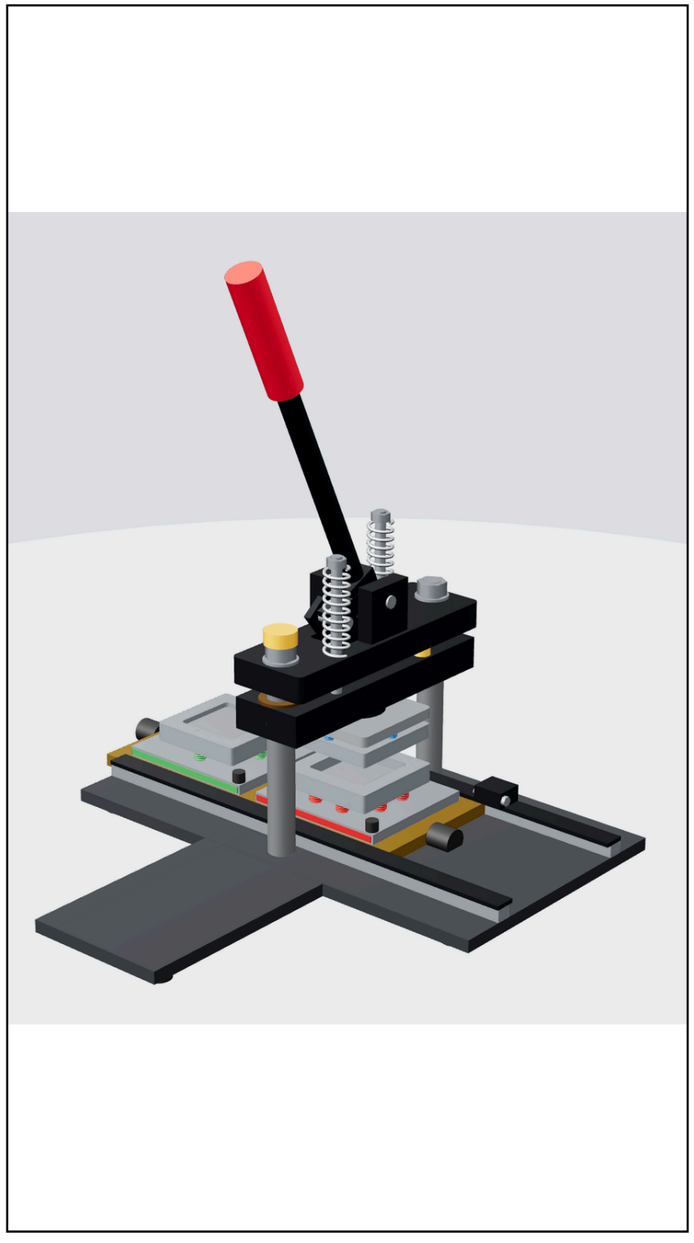
MagPress M4

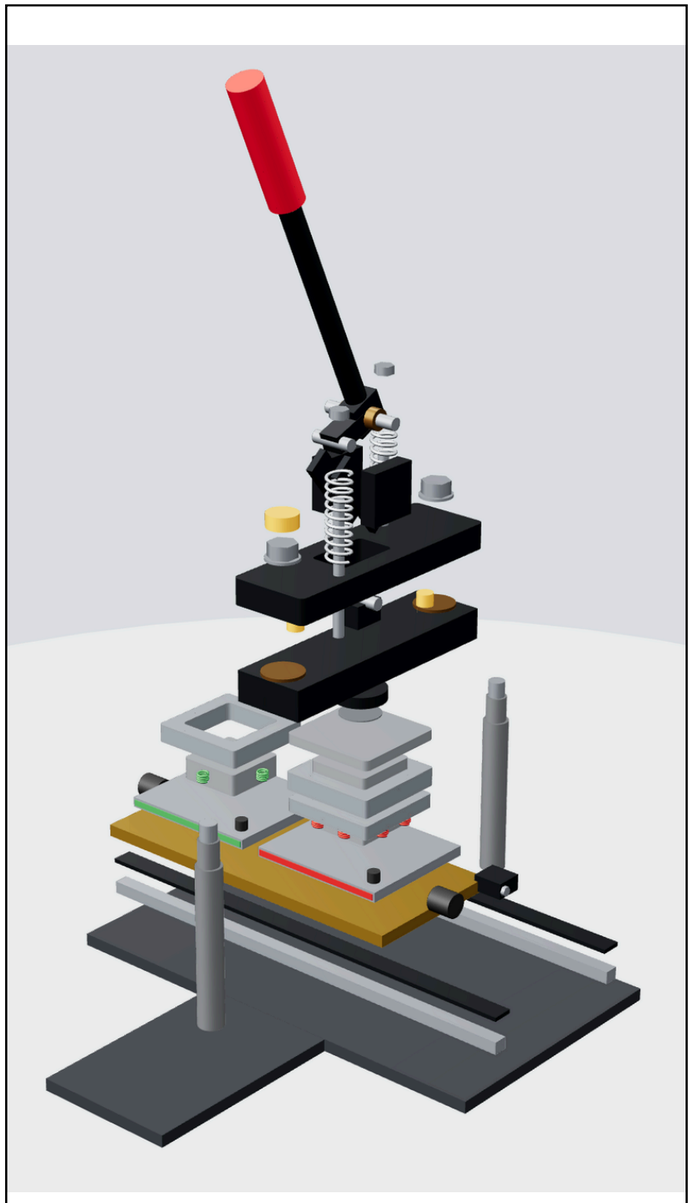
Manual photo fridge-magnet press · manufacturing drawing set · Rev A

4 swappable die sets: 50 × 50, 63.5 × 63.5, 53 × 80, 65 × 90 mm

DRAWING LIST

DRG NO.	TITLE	QTY	SHEET
MP-001	GENERAL ARRANGEMENT & HEIGHTS	—	2
MP-101	BASE PLATE	1	3
MP-102	GUIDE RAIL	2 (1 × -F, 1 × -B)	4
MP-103	KEEPER STRIP (GIB)	2 (1 × -F, 1 × -B)	5
MP-104	STATION STOP BLOCK	2	6
MP-105	DIE TABLE (SLIDE)	1	7
MP-106	DIE TABLE STOP PIN	1	8
MP-107	COLUMN	2	9
MP-108	CROWN	1	10
MP-109	RAM	1	11
MP-110	CLEVIS	1	12
MP-111	DIE HOLDER	1	13
MP-112	PIVOT CHEEK	2	14
MP-113	HANDLE HUB + CRANK	1	15
MP-114	TOGGLE LINK	2	16
MP-115	HANDLE BAR	1	17
MP-116	PIN A / PIN B (TOGGLE)	2	18
MP-117	PIN O (HANDLE PIVOT)	1	19
MP-118	RETURN ROD	2	20
MP-119	PIVOT SPACER	2	21
MP-201	UPPER DIE BODY	1 per die set (4)	22
MP-202	UPPER DIE SHANK	1 per die set (4)	23
MP-203	UPPER DIE PLUG	1 per die set (4)	24
MP-204	UPPER DIE FRAME	1 per die set (4)	25
MP-205	LOWER DIE SHOE A / B	2 per set (A + B)	26
MP-206	PICKUP PEDESTAL (DIE A)	1 per die set (4)	27
MP-207	PICKUP FRAME (DIE A)	1 per die set (4)	28
MP-208	CRIMP PEDESTAL (DIE B)	1 per die set (4)	29
MP-209	CRIMP FRAME (DIE B)	1 per die set (4)	30





Reference renderings: assembled, handle up (left); exploded (right). Not for measurement.

NOTES

1. Units mm. First-angle projection (ISO 128-30); every view is also labelled by name.
2. General tolerances ISO 2768-mK unless stated. Fits ISO 286. Threads ISO metric coarse 6H/6g unless stated.
3. Surface roughness Ra 3.2 unless stated. Break sharp edges 0.3 × 45°; forming edges as drawn.
4. Column bores in base MP-101, crown MP-108 and ram MP-109 at 210.00 ±0.01: same datum scheme and CNC program.
5. Die drawings MP-201 to MP-209 are tabulated for four sizes (5050, 6363, 5380, 6590). Make one part per size.
6. DXF profiles are supplied for cutting and wire-EDM. If a DXF and a drawing disagree, the drawing governs.
7. Bought parts (springs, screws, bushings, dowels, grips, buffers) are listed in the build pack BOM.
8. Rev A: first issue for prototype. Prove the die fits on a 7075 aluminium set before cutting tool steel.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **A**	SHEET: 1 / 30

TITLE

DRAWING SET — COVER & LIST

DRAWING NO.	QTY	SCALE	UNITS
MP-000	—	—	mm

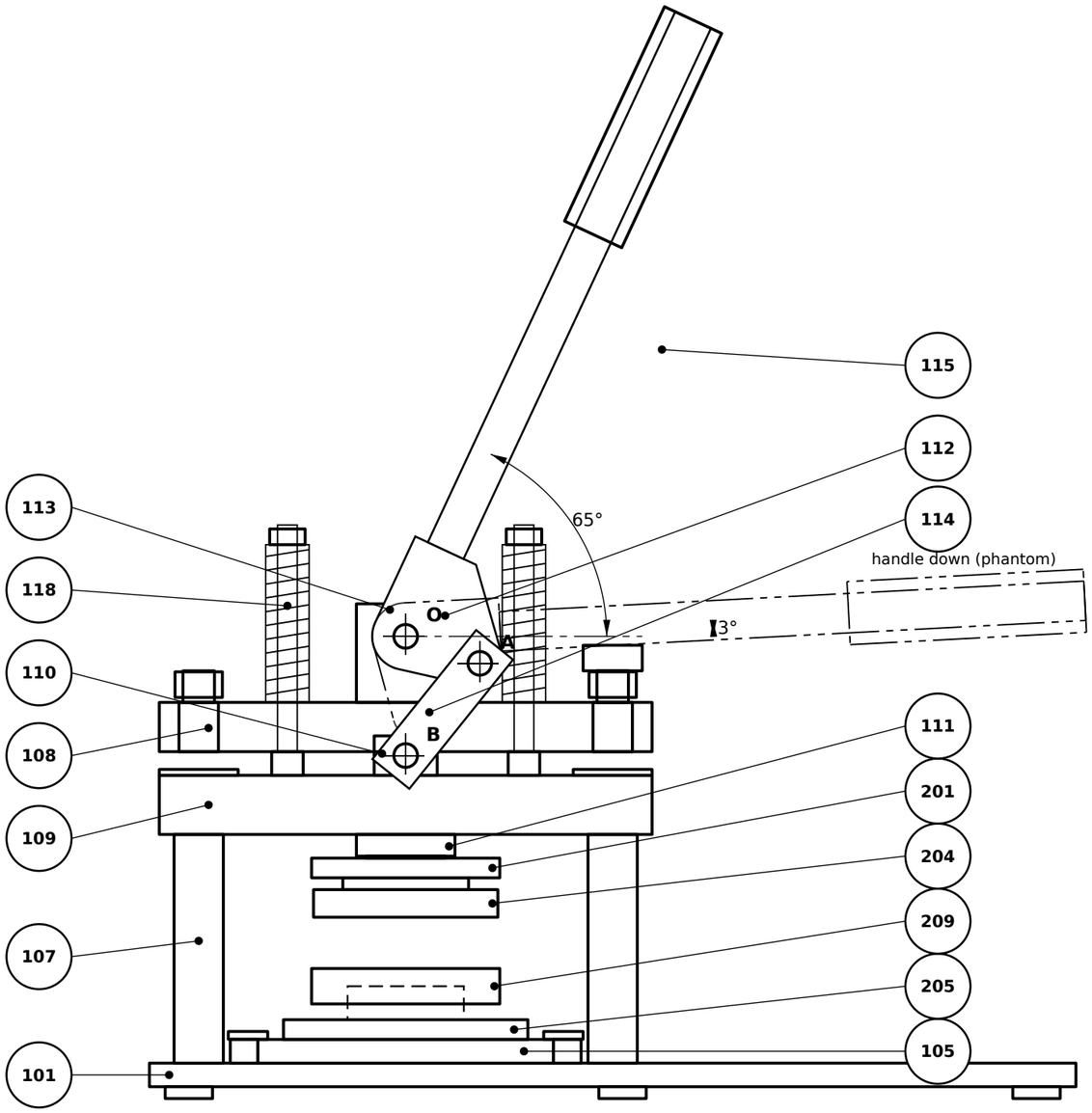
MATERIAL: —

HEAT TREATMENT / FINISH: —

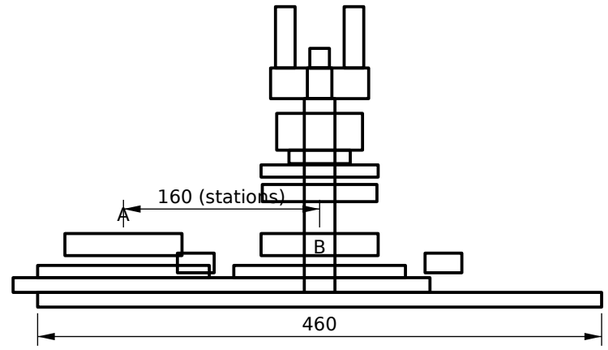
GENERAL TOLERANCES		DATE	DRAWN	CHECKED
ISO 2768-mK	1st	2026-10-26		

KEY HEIGHTS, handle up (mm from bench)

HEIGHT	LEVEL
234.5	pivot O
201	crown top
176	crown bottom
164	ram top
134	ram bottom
92	upper die bottom
66	lower die top
40	shoe top
30	die table top
18	base top
0	bench



SIDE ELEVATION (viewed from the left; front →)



FRONT ELEVATION 1:5

NOTES

1. Shown with the handle UP (open), die table at "B under ram", die set 6363. Heights from the bench top, handle up.
2. Toggle: crank O→A 40, link A→B 60, grip centre 290 from O. Handle travel 65° → 3° above horizontal; crank 70° → 8° from vertical.
3. Ram stroke 38.9. Upper die meets lower die after 26.0; die working travel 12.9. Stroke 1 (pickup) stops at 35.5 when the plug meets the shell.
4. Open stop: 2 urethane bumpers (ram top) against the crown. Closed stop: urethane buffer on the front column top under the bar.
5. Die table stations at X = ±80 from table centre (travel 160). Stop blocks MP-104 set each station under the ram centre ±0.05.
6. Balloons = drawing numbers (MP-xxx). Pins MP-116 at A and B, MP-117 at O; spacers MP-119 on pin O; stop blocks MP-104 and stop pin MP-106 behind the table.

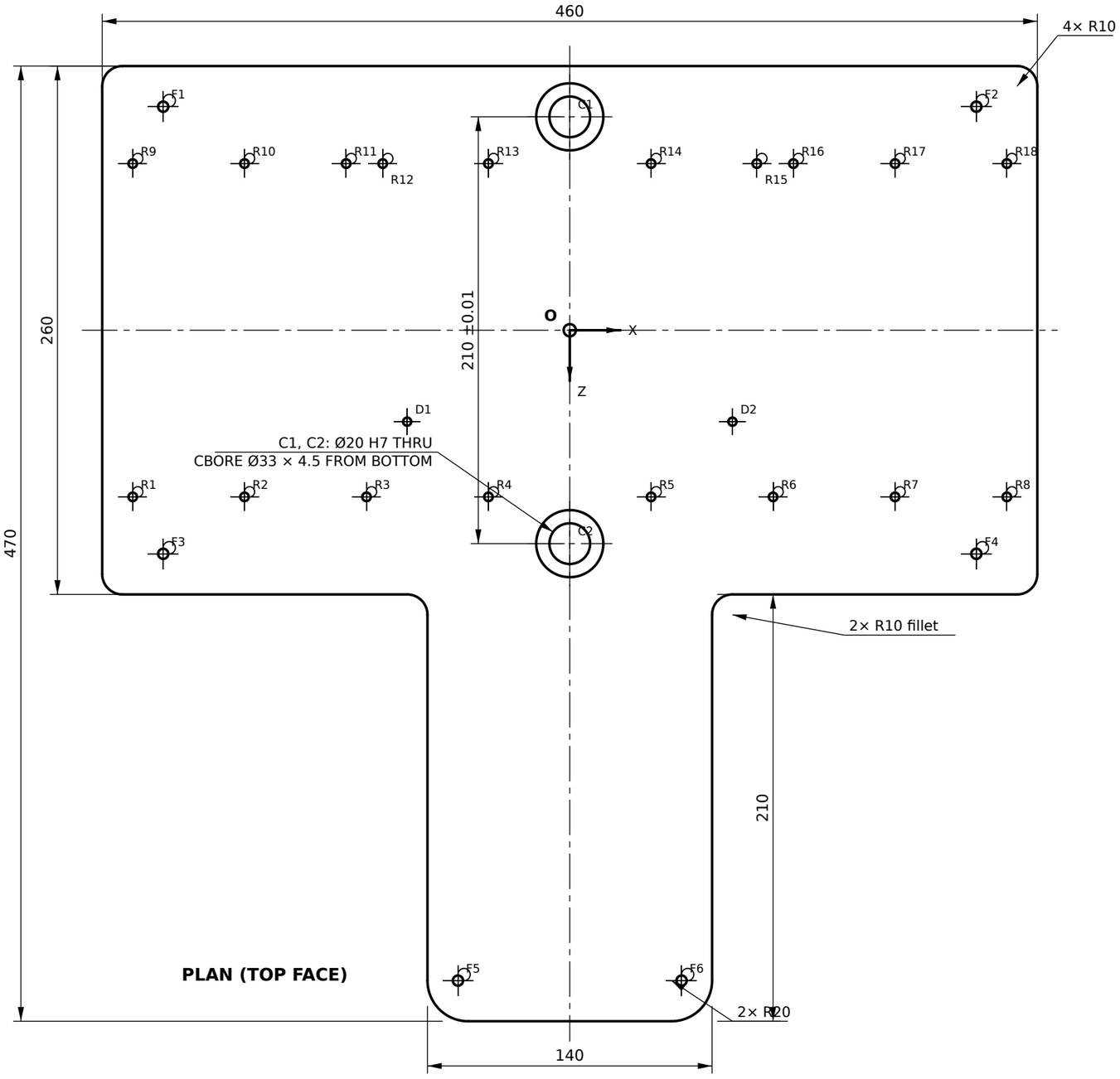
PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **A**	SHEET: 2 / 30

TITLE

GENERAL ARRANGEMENT & HEIGHTS

DRAWING NO.	QTY	SCALE	UNITS
MP-001	—	1:3 / 1:5	mm

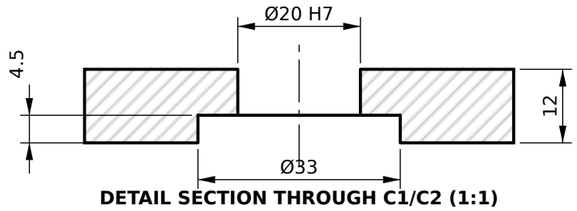
MATERIAL: —				
HEAT TREATMENT / FINISH: —				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



PLAN (TOP FACE)

HOLE TABLE

HOLE	X	Z	DESCRIPTION
C1	0	-105	Ø20 H7 THRU; CBORE Ø33 × 4.5 BOTTOM
C2	0	105	Ø20 H7 THRU; CBORE Ø33 × 4.5 BOTTOM
R1	-215	82	M5 THRU (rail screws)
R2	-160	82	M5 THRU (rail screws)
R3	-100	82	M5 THRU (rail screws)
R4	-40	82	M5 THRU (rail screws)
R5	40	82	M5 THRU (rail screws)
R6	100	82	M5 THRU (rail screws)
R7	160	82	M5 THRU (rail screws)
R8	215	82	M5 THRU (rail screws)
R9	-215	-82	M5 THRU (rail screws)
R10	-160	-82	M5 THRU (rail screws)
R11	-110	-82	M5 THRU (rail screws)
R12	-92	-82	M5 THRU (rail screws)
R13	-40	-82	M5 THRU (rail screws)
R14	40	-82	M5 THRU (rail screws)
R15	92	-82	M5 THRU (rail screws)
R16	110	-82	M5 THRU (rail screws)
R17	160	-82	M5 THRU (rail screws)
R18	215	-82	M5 THRU (rail screws)
F1	-200	-110	M6 THRU (feet)
F2	200	-110	M6 THRU (feet)
F3	-200	110	M6 THRU (feet)
F4	200	110	M6 THRU (feet)
F5	-55	320	M6 THRU (feet)
F6	55	320	M6 THRU (feet)
D1	-80	45	Ø4 × 90° spot, 1.5 deep (detent)
D2	80	45	Ø4 × 90° spot, 1.5 deep (detent)



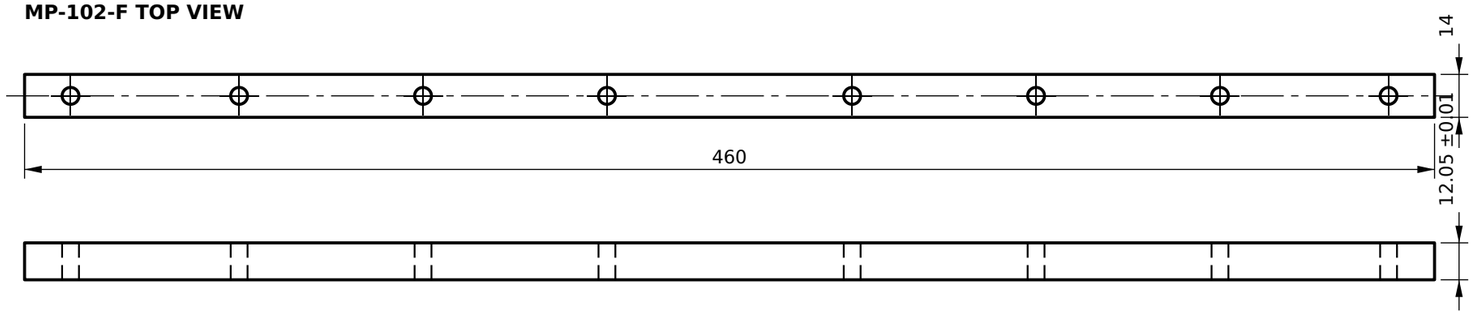
DETAIL SECTION THROUGH C1/C2 (1:1)

NOTES

1. Blank: profile-cut 16 mm S235JR. Face both sides to 12.0 ±0.1. Flatness 0.05 over slide area X ±230 × Z ±75.
2. Top face in slide area Ra 1.6 and left unpainted (oil). The die table slides directly on it.
3. C1/C2 Ø20 H7 at 210.00 ±0.01 pitch: machine with the same datum and program as crown MP-108 and ram MP-109.
4. Hole positions from datum O (midpoint of C1–C2). X to the right; Z toward the front (tongue).
5. C1/C2: counterbore Ø33 × 4.5 deep from the BOTTOM for the column clamp washer.
6. D1/D2: Ø4 × 90° countersink spot 1.5 deep, top face (ball-plunger detents for die table stations).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 3 / 30
TITLE: **BASE PLATE**			
DRAWING NO.: **MP-101**	QTY: 1	SCALE: 1:2.5	UNITS: mm
MATERIAL: S235JR steel plate 16 mm, profile cut, faced to 12.0			
HEAT TREATMENT / FINISH: Black powder coat 60–80 µm; mask top face inside rails, bores, threads, counterbores			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:

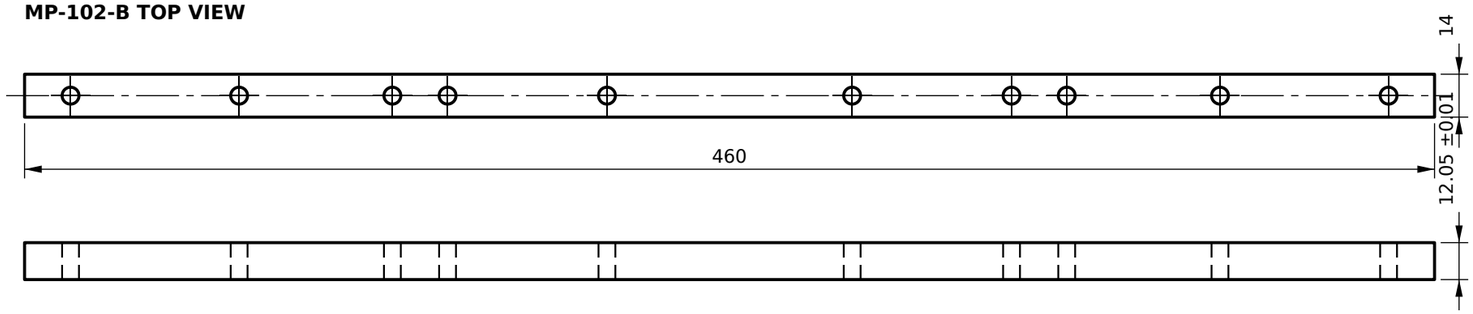
MP-102-F TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

-215, -160, -100, -40, 40, 100, 160, 215

MP-102-B TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

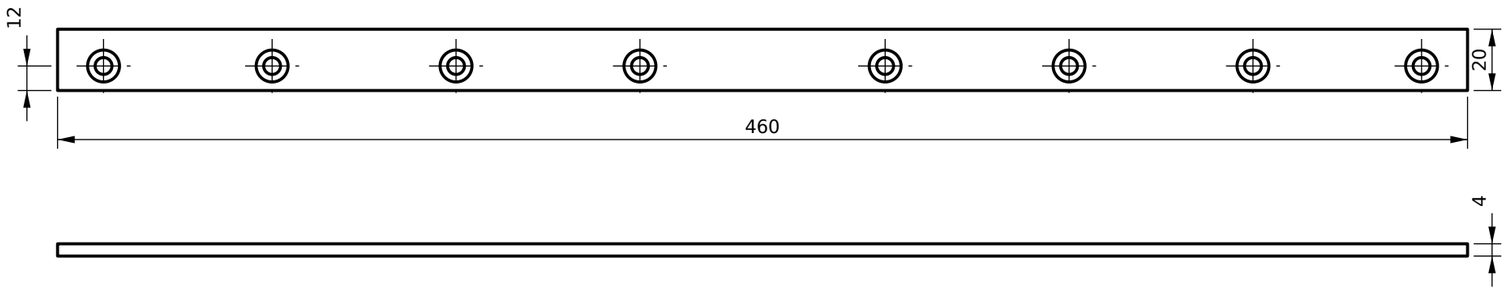
-215, -160, -110, -92, -40, 40, 92, 110, 160, 215

NOTES

1. Height 12.05 ±0.01: equal to die table MP-105 thickness + 0.05 running clearance. Grind both rails together.
2. Ground faces: top, bottom and inner face (faces the die table). Inner face straight within 0.02 over 460.
3. Variant -F: front rail, 8 holes. Variant -B: back rail, 10 holes (includes stop-block screws).
4. Holes Ø5.5 THRU on the rail centreline; hole X positions measured from the rail centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 4 / 30
TITLE: **GUIDE RAIL**			
DRAWING NO.: **MP-102**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 16 × 14, ground to 14 × 12.05			
HEAT TREATMENT / FINISH: Black oxide; ground faces oiled			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

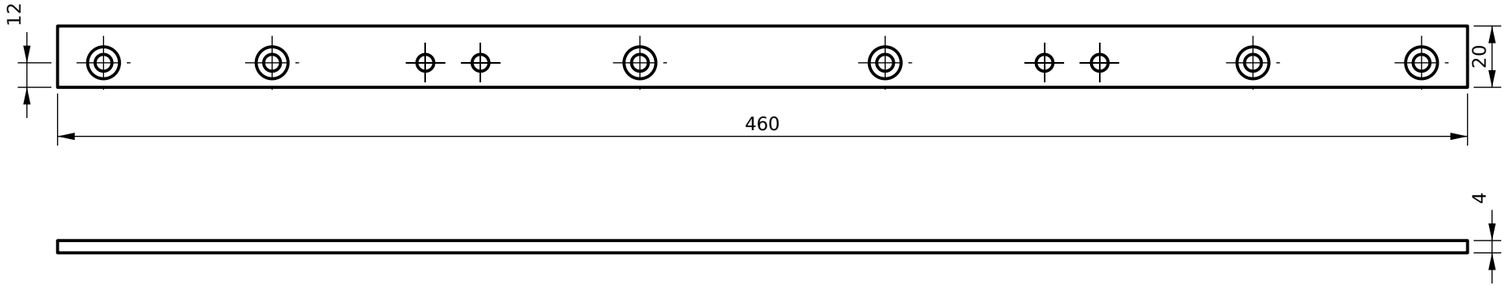
MP-103-F TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -100, -40, 40, 100, 160, 215

MP-103-B TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -40, 40, 160, 215

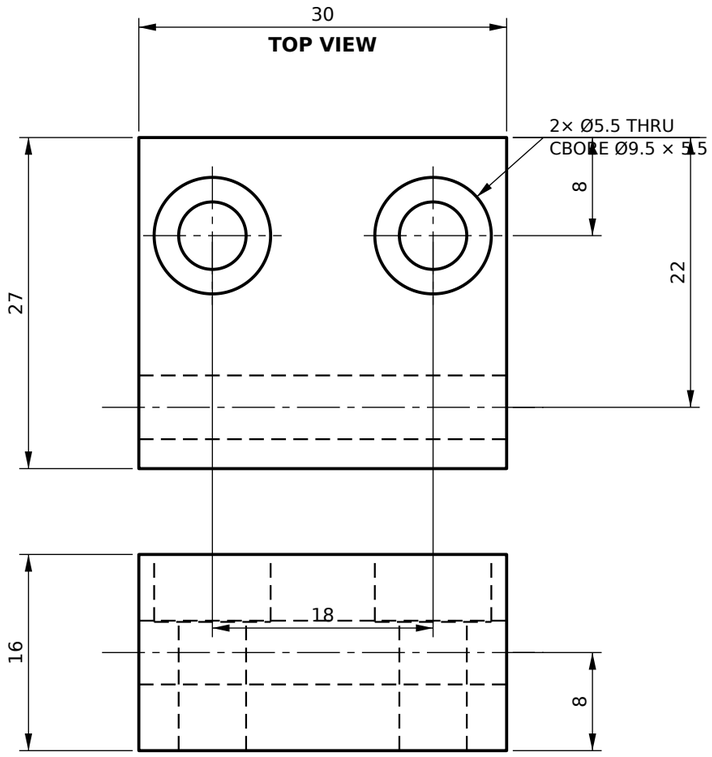
Plain Ø5.5 THRU:

-110, -92, 92, 110

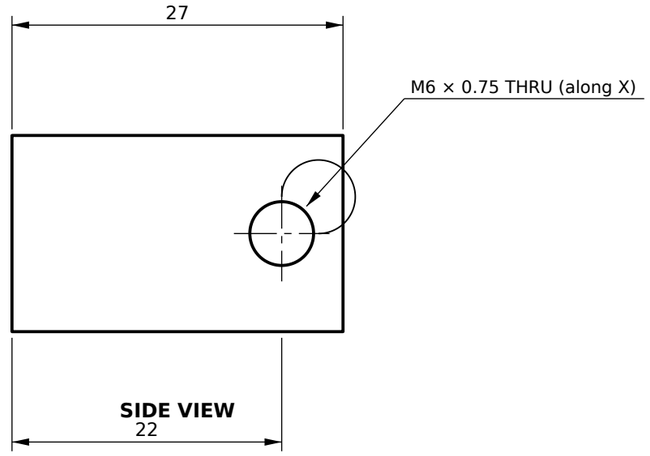
NOTES

1. Sits on top of the guide rail and overlaps the die table edge by 5 mm (inner edge).
2. Countersunk holes: Ø5.5 THRU, 90° countersink to Ø10.4 (for M5 ISO 10642), from top.
3. Variant -B: four plain Ø5.5 holes at X ±92 and ±110 for the stop-block screws (no countersink).
4. Hole line 12.0 from the inner edge. Hole X positions measured from the strip centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 5 / 30
TITLE: **KEEPER STRIP (GIB)**			
DRAWING NO.: **MP-103**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 20 × 4			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



FRONT VIEW (from the die table side)

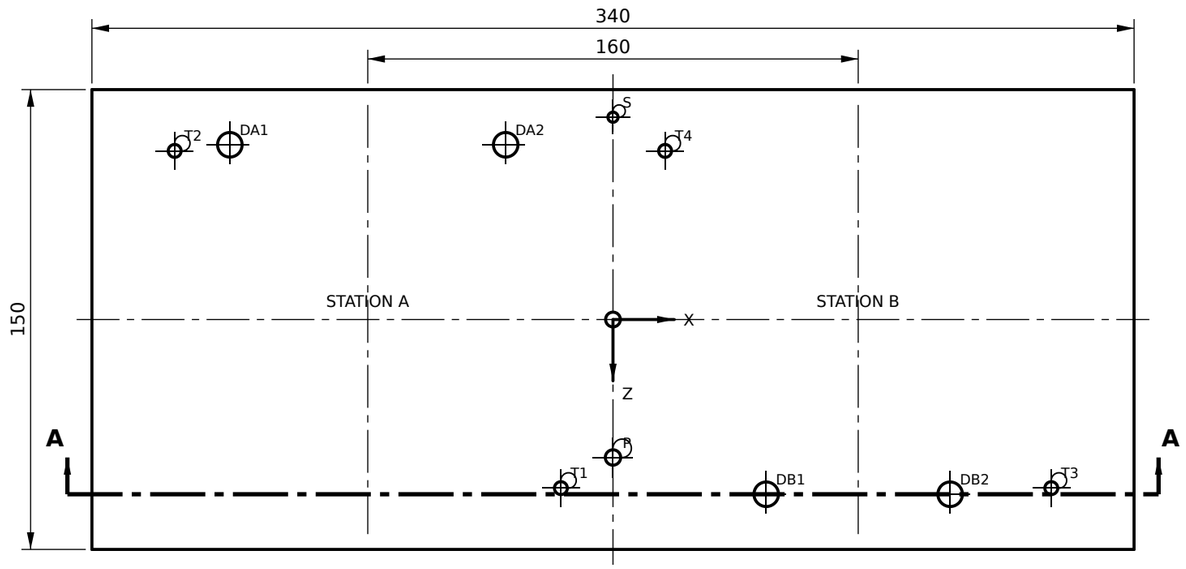


SIDE VIEW

NOTES

1. Bolted through keeper MP-103-B and rail MP-102-B into the base with 2 × M5 × 35 ISO 4762.
2. Stop screw: M6 × 0.75 × 25 hex screw + thin lock nut (bought). Nominal protrusion 3 mm from the inner face.
3. The 7 mm overhang (Z 20–27) sits above the die table; the table stop pin MP-106 hits the screw tip.
4. Z measured from the back face (the face over the keeper).
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

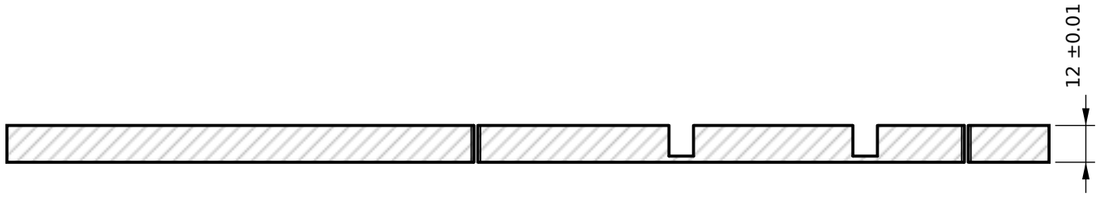
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 6 / 30
TITLE: **STATION STOP BLOCK**			
DRAWING NO.: **MP-104**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 bright 30 × 16 flat			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



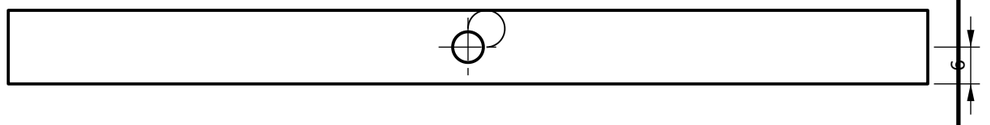
PLAN

HOLE TABLE

HOLE	X	Z	DESCRIPTION
DA1	-125	-57	Ø8 H7 × 10 deep (shoe dowel)
DA2	-35	-57	Ø8 H7 × 10 deep (shoe dowel)
DB1	50	57	Ø8 H7 × 10 deep (shoe dowel)
DB2	110	57	Ø8 H7 × 10 deep (shoe dowel)
T1	-17	55	STI M5 THRU for Helicoil (thumb screw)
T2	-143	-55	STI M5 THRU for Helicoil (thumb screw)
T3	143	55	STI M5 THRU for Helicoil (thumb screw)
T4	17	-55	STI M5 THRU for Helicoil (thumb screw)
S	0	-66	M4 × 8 deep (stop pin MP-106)
P	0	45	M6 THRU (ball plunger)
K1	-170	0	M6 × 12 in end face, Y = 6 (knob)
K2	170	0	M6 × 12 in end face, Y = 6 (knob)



SECTION A–A

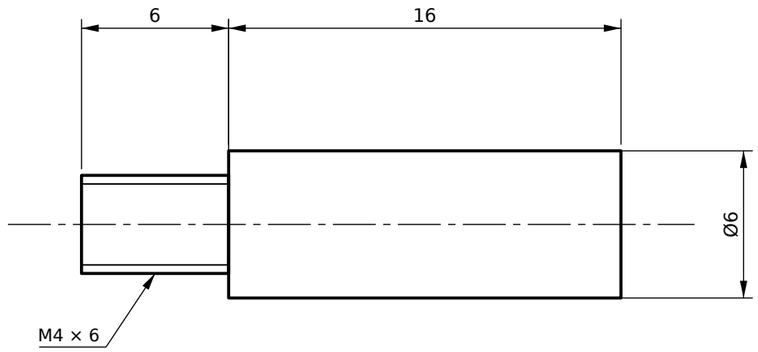


END VIEW (K1/K2) 1:1

NOTES

1. Thickness 12.00 ±0.01 and flat within 0.03 after anodising. Guide rails MP-102 are 12.05.
2. Station A dowels DA1/DA2 lie on a line parallel to X; station B dowels DB1/DB2 use a different pitch so the A and B die shoes cannot be swapped.
3. DA/DB: Ø8 H7 × 10 deep (after anodise). T1–T4: tap STI M5 for M5 × 1D Helicoil, THRU. S: M4 × 8 for stop pin MP-106.
4. P: M6 THRU for M6 spring ball plunger (detents D1/D2 in the base). K1/K2: M6 × 12 in each end face, centred, for push knobs.
5. Hole positions from the table centre. Station centres at X = −80 (A) and +80 (B), Z = 0.
6. Long edges: 0.5 × 45° chamfer only (they run against the rails).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **A**	SHEET: 7 / 30	
TITLE: **DIE TABLE (SLIDE)**			
DRAWING NO.: **MP-105**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: EN AW-6061-T6 plate 15 mm, faced to 12.00			
HEAT TREATMENT / FINISH: Hard anodise 25 µm, gold; fit Helicoils after anodising			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

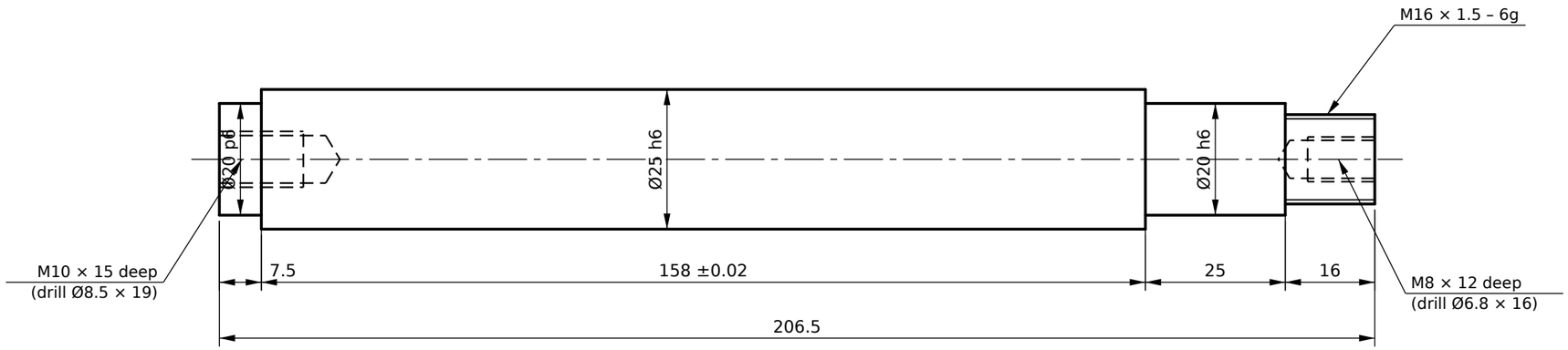


SIDE VIEW

NOTES

1. Screwed into the die table (MP-105, hole S) with Loctite 243.
2. Contact face Ø6 must be square to the axis within 0.05.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 8 / 30
TITLE: **DIE TABLE STOP PIN**			
DRAWING NO.: **MP-106**	QTY: 1	SCALE: 4:1	UNITS: mm
MATERIAL: C45 Ø8 bright bar			
HEAT TREATMENT / FINISH: Harden 45–50 HRC; black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

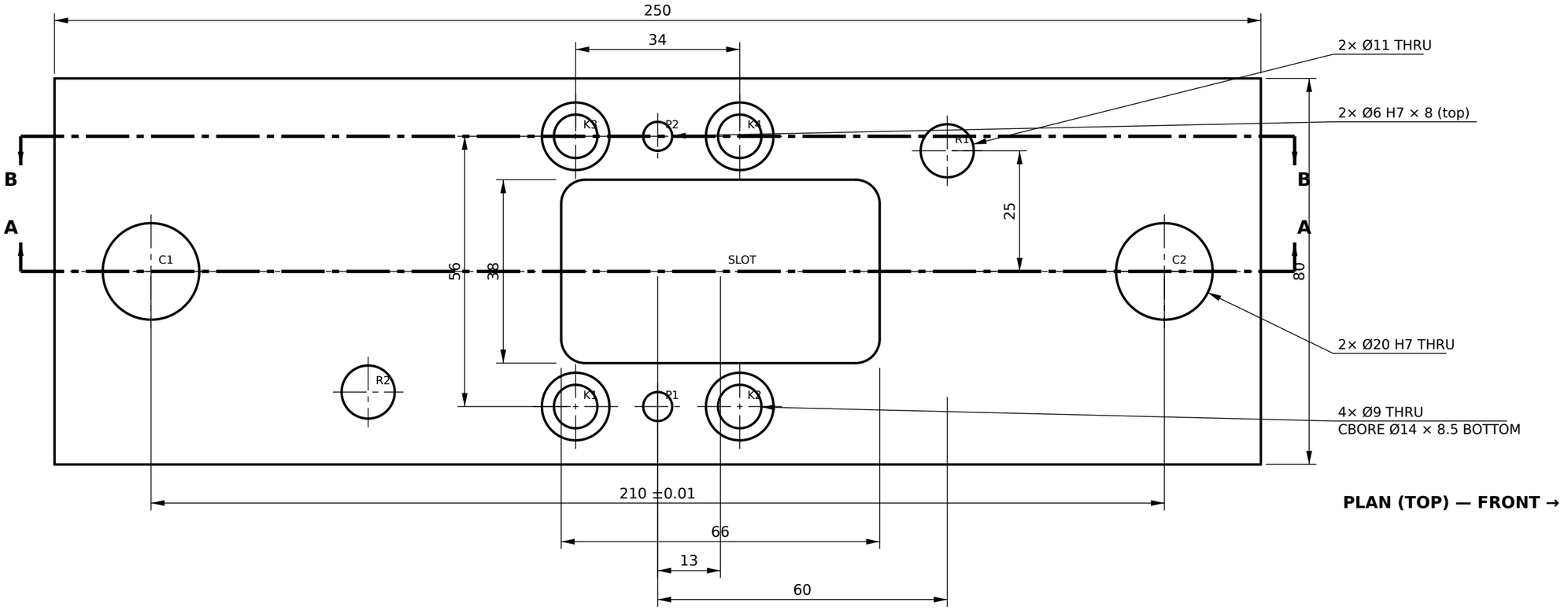


SIDE VIEW

NOTES

1. Ø25 h6 length 158.00 ±0.02; the two columns matched within 0.01 (sets crown parallelism).
2. Ø20 p6 bottom spigot: press into base MP-101 (Ø20 H7). Clamp from below: M10 × 25 ISO 4762 + washer Ø32 × 4.
3. Ø20 h6 top spigot passes through the crown MP-108 (Ø20 H7); M16 × 1.5 nut and hardened washer on top, 100 N·m.
4. Top end M8 × 12: front column takes the urethane handle-stop buffer Ø30 × 13 (bought).
5. Shoulders square to axis within 0.01. Run-out of all diameters 0.01 to the Ø25 axis.
6. Chamfers: ends 1 × 45°; steps 0.5 × 45°. Thread undercut DIN 76-A at the M16 thread end.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

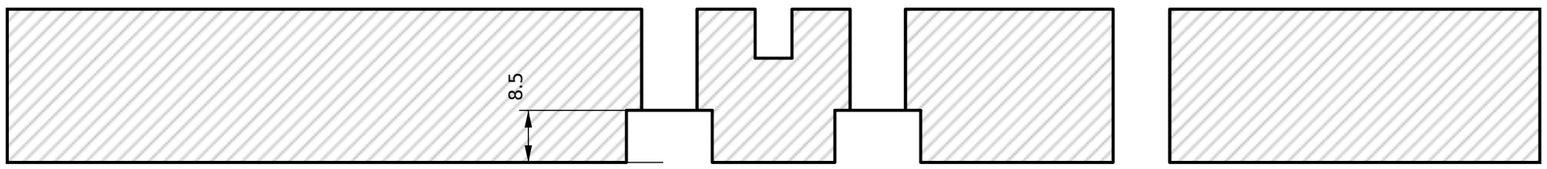
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 9 / 30
TITLE: **COLUMN**			
DRAWING NO.: **MP-107**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: 100Cr6 (52100 / GCr15) Ø30 bar			
HEAT TREATMENT / FINISH: Ø25 section: induction harden 58–62 HRC, hard chrome 8–12 µm, ground to h6, Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



PLAN (TOP) — FRONT →



SECTION A–A

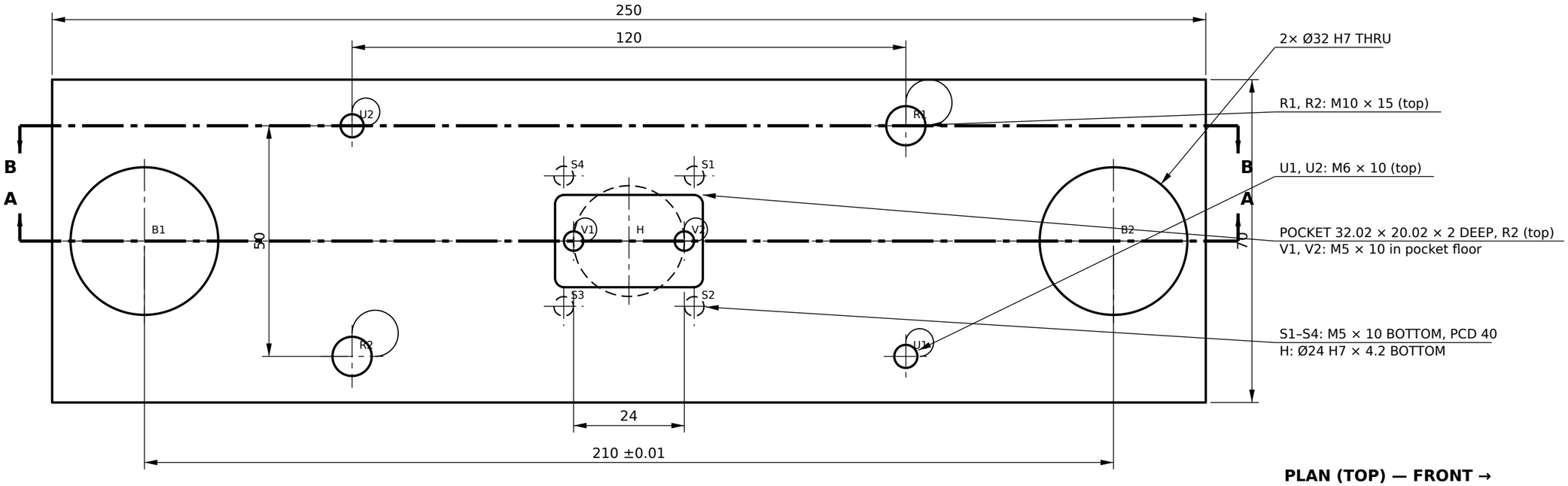


SECTION B–B (X = +28)

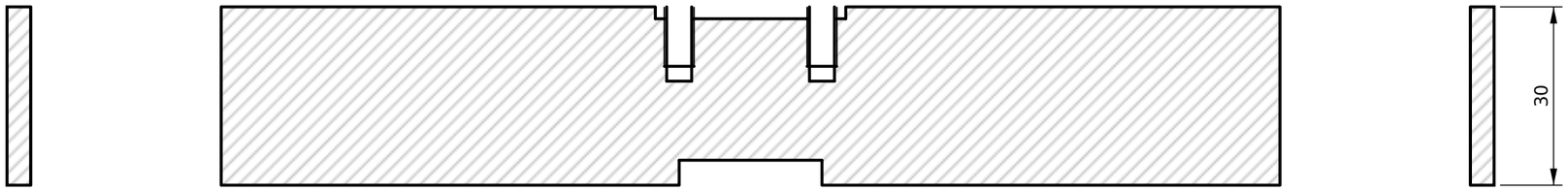
NOTES

1. C1/C2 Ø20 H7 at 210.00 ±0.01: same datum and program as base MP-101 and ram MP-109.
2. Top and bottom faces parallel 0.02, flat 0.02. Bores square to the bottom face within 0.01/25.
3. K1–K4: Ø9 THRU, counterbore Ø14 × 8.5 from the BOTTOM (M8 × 30 ISO 4762 into the pivot cheeks MP-112).
4. P1/P2: Ø6 H7 × 8 deep from the top (cheek dowels). Slot 38 × 66, R5 corners, THRU (links pass through).
5. Plan drawn with the front of the press to the right. X positive = the operator's right.
6. Check at 8 kN: bending stress ≈ 96 MPa, centre deflection ≈ 0.13 mm.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

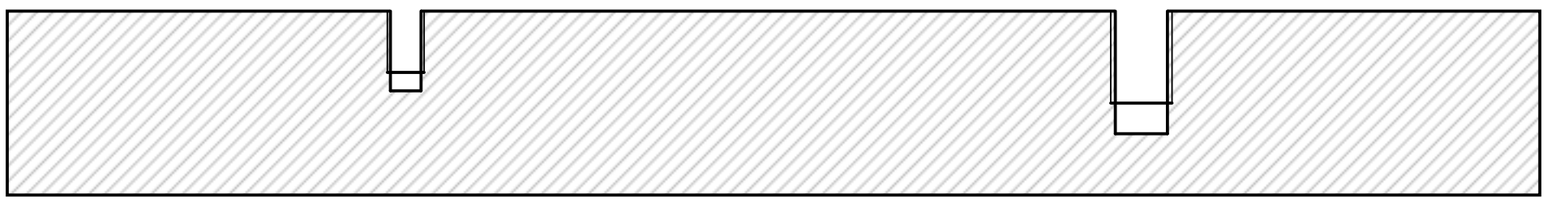
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 10 / 30
TITLE: **CROWN**			
DRAWING NO.: **MP-108**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 80 × 30, faced to 25			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



PLAN (TOP) — FRONT →



SECTION A–A

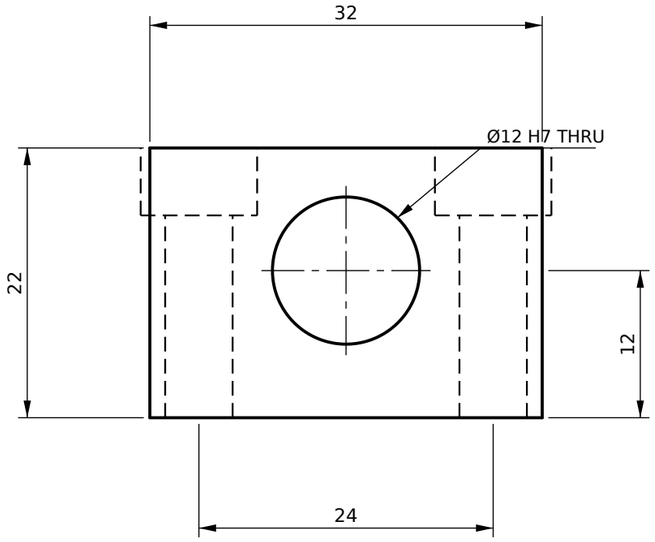


SECTION B–B (X = +25)

NOTES

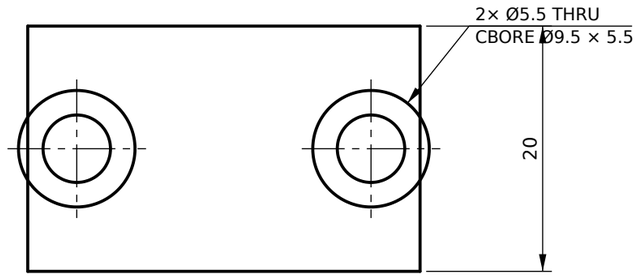
1. B1/B2 Ø32 H7 at 210.00 ±0.01 for bushings (self-lubricating bronze, ID 25 / OD 32 / L 33 flanged): same program as MP-101/MP-108.
2. Top: clevis pocket 20.02 × 32.02 × 2.0 deep (R2 corners) with V1/V2 M5 × 10. R1/R2 M10 × 15 (return rods). U1/U2 M6 × 10 (urethane bumpers Ø16 × 12).
3. Bottom: H = Ø24 H7 × 4.2 deep (die-holder spigot), S1–S4 M5 × 10 on PCD 40 at 45°.
4. Top and bottom faces parallel 0.02. Bores square to faces within 0.01/30.
5. Plan drawn with the front of the press to the right. Hidden features are on the bottom face.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 11 / 30
TITLE: **RAM**			
DRAWING NO.: **MP-109**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 80 × 35, faced to 70 × 30			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



FRONT VIEW (pin axis = X)

TOP VIEW

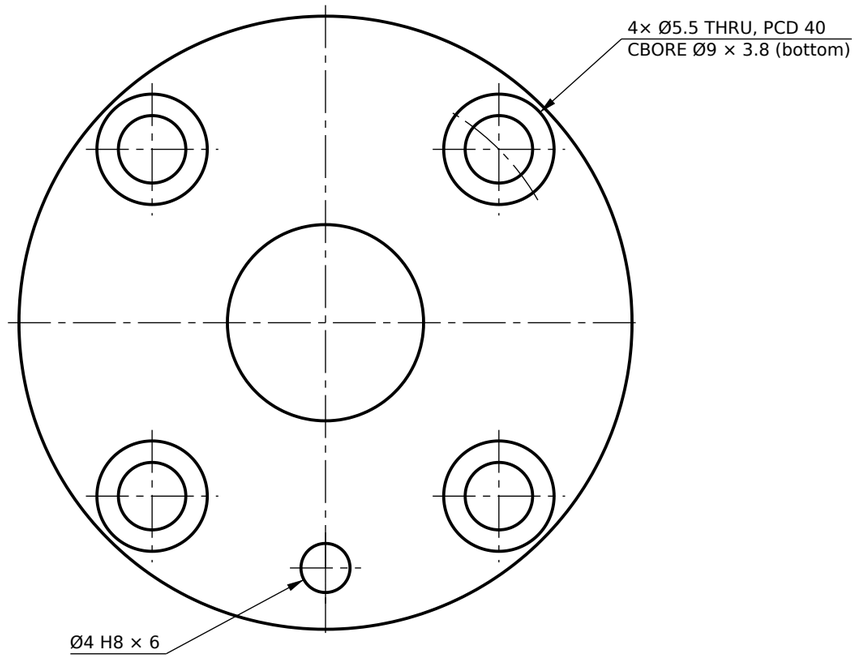


NOTES

1. Sits 2 mm deep in the ram pocket; 2 × M5 × 25 ISO 4762. Pin B Ø12 g6 runs in the Ø12 H7 bore.
2. Pin bore axis parallel to the bottom face within 0.02 and square to the long sides within 0.02.
3. Vertical edges chamfer 2.5 × 45° to clear the R2 pocket corners.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 12 / 30
TITLE: **CLEVIS**			
DRAWING NO.: **MP-110**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 25 × 35			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

SECTION THROUGH X AXIS

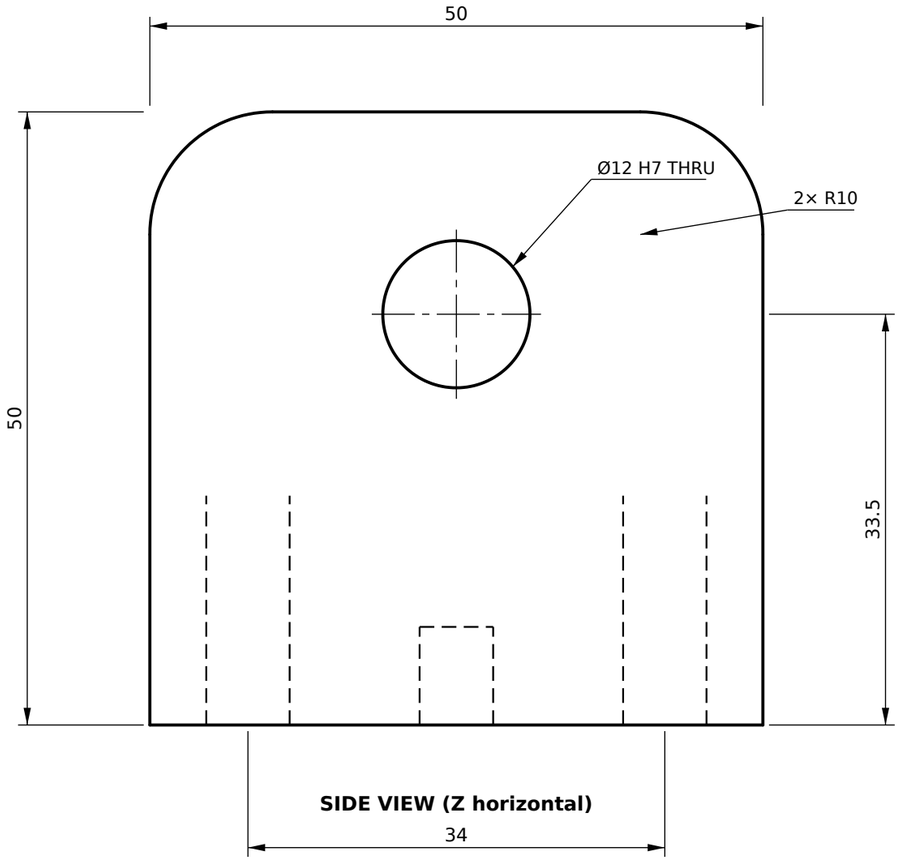


BOTTOM VIEW (front at bottom)

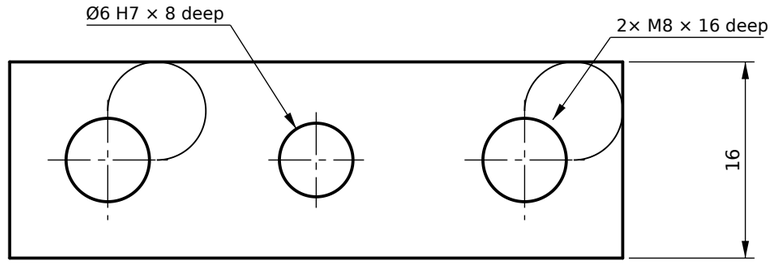
NOTES

1. Spigot Ø24 h6 locates in the ram (Ø24 H7). 4 × M5 × 16 DIN 7984 low-head screws from below.
2. Bore Ø16 H7 takes the upper-die shank MP-202. Bottom face (die seat) square to the bore within 0.01, flat 0.01.
3. Lock screw: M5 × 10 ISO 4027 cone-point set screw in the radial M5 hole; it seats in the shank V-groove.
4. Orientation hole Ø4 H8 × 6 deep in the bottom face at 20 mm radius, on the Z axis toward the FRONT.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 13 / 30
TITLE: **DIE HOLDER**			
DRAWING NO.: **MP-111**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 42CrMo4 Ø55 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



SIDE VIEW (Z horizontal)

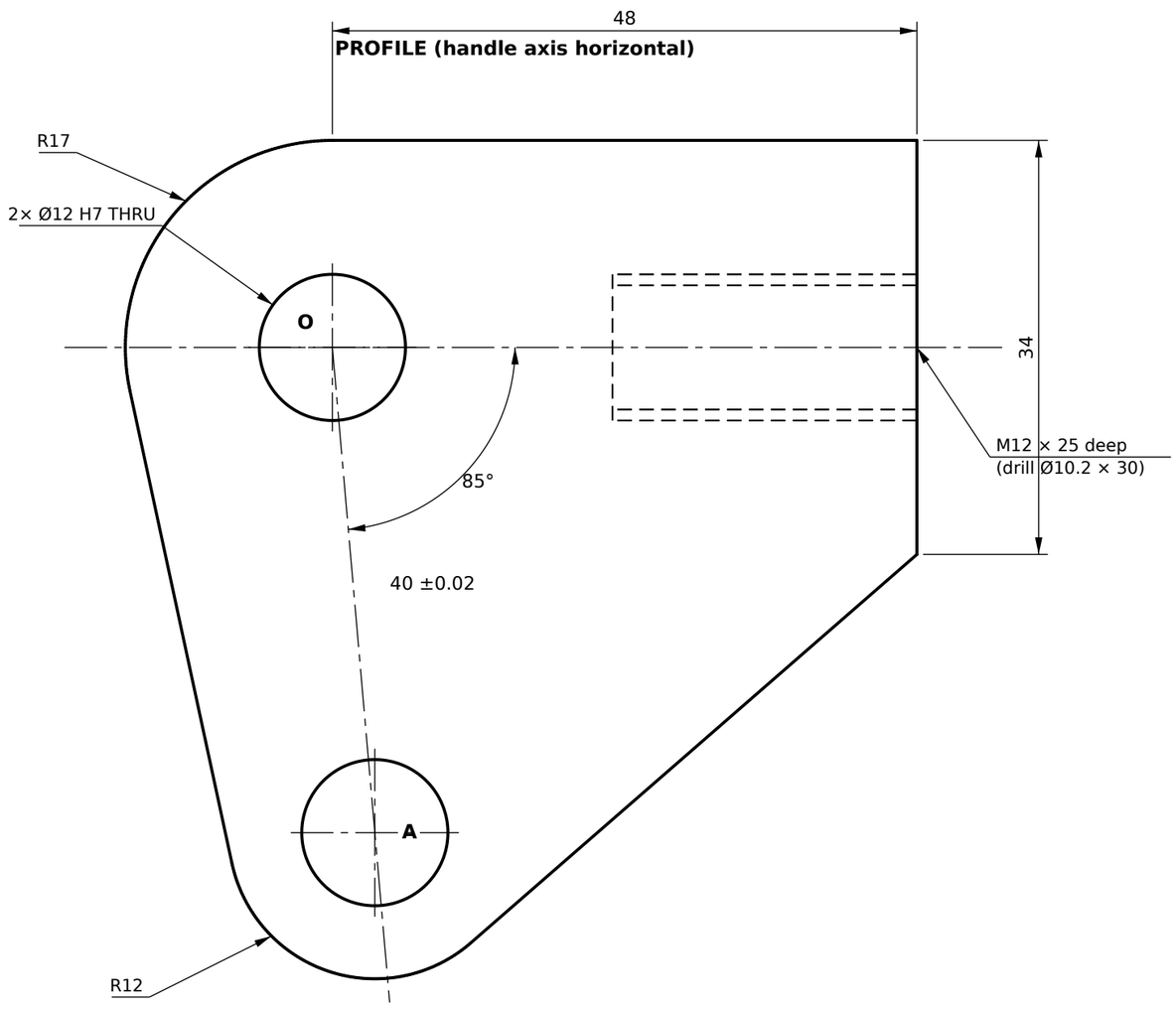


BOTTOM VIEW

NOTES

1. Ø12 H7 pivot bores of both cheeks line-reamed together after fitting to the crown (pin O must slide in freely).
2. Fixed to the crown with 2 × M8 × 30 ISO 4762 from below + 1 × Ø6 × 16 dowel. Carries the full toggle reaction (8 kN design).
3. Bottom face flat 0.02, square to the sides.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

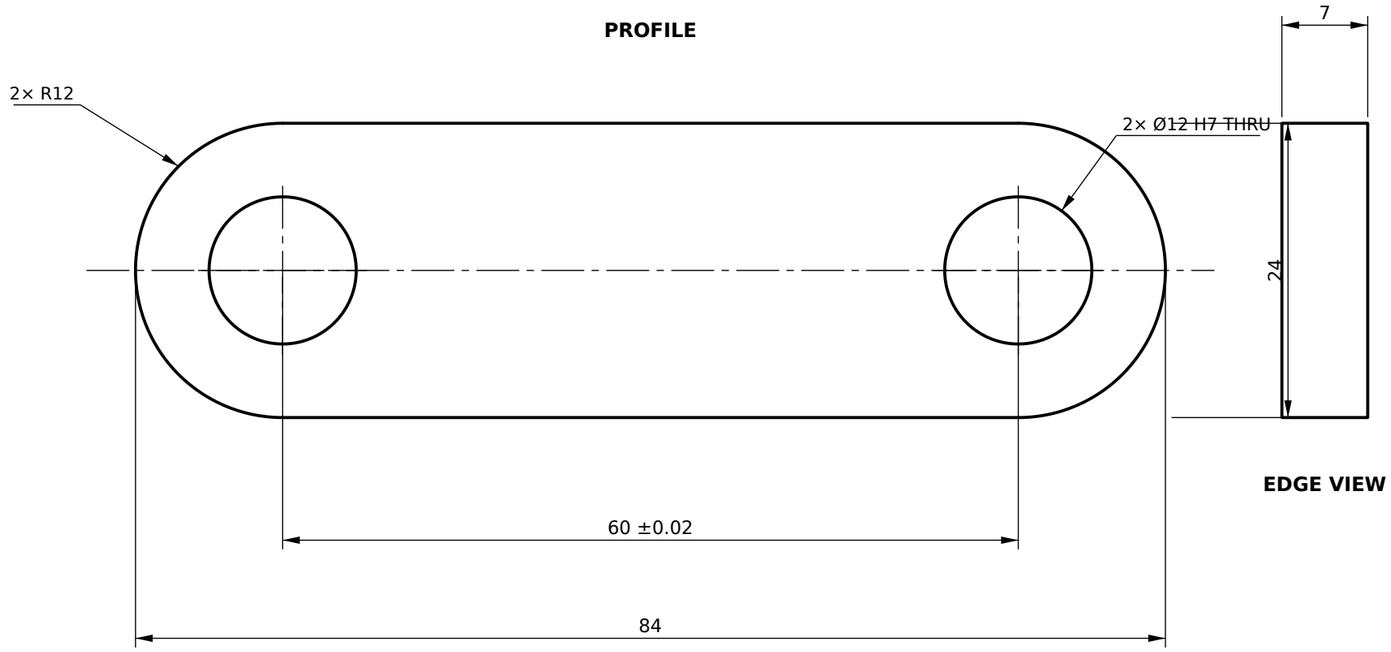
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 14 / 30
TITLE: **PIVOT CHEEK**			
DRAWING NO.: **MP-112**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 20 × 55			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



NOTES

1. Profile from DXF (laser/waterjet + machine, or wire EDM). Faces parallel 0.02.
2. Crank radius O→A = 40.00 ±0.02 at 85° ±0.1° from the handle axis. Ø12 H7 bores parallel within 0.02/20.
3. M12 × 25 deep tapped along the handle axis from the end face (handle bar MP-115). End face square to the axis 0.05.
4. O runs on pin O (MP-117) between two spacers MP-119; A carries pin A and both toggle links MP-114.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

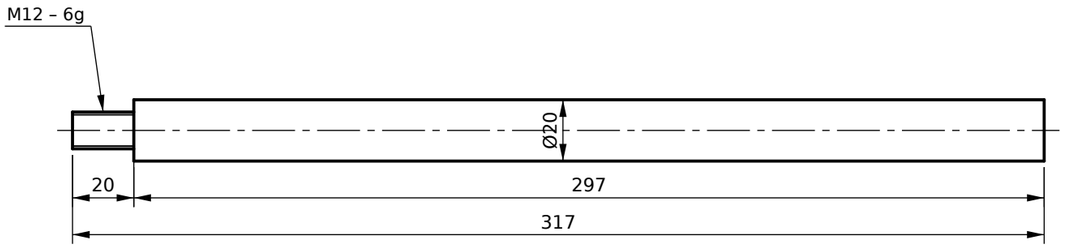
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 15 / 30
TITLE: **HANDLE HUB + CRANK**			
DRAWING NO.: **MP-113**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: C45 plate 25 mm, faced to 20			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:



NOTES

1. Hole centres 60.00 ±0.02; make both links as a matched pair (within 0.01). Bores Ø12 H7 with an oil groove.
2. Faces parallel 0.02. Stamp the pair with matching numbers.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 16 / 30
TITLE: **TOGGLE LINK**			
DRAWING NO.: **MP-114**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 plate 8 mm, faced to 7			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:

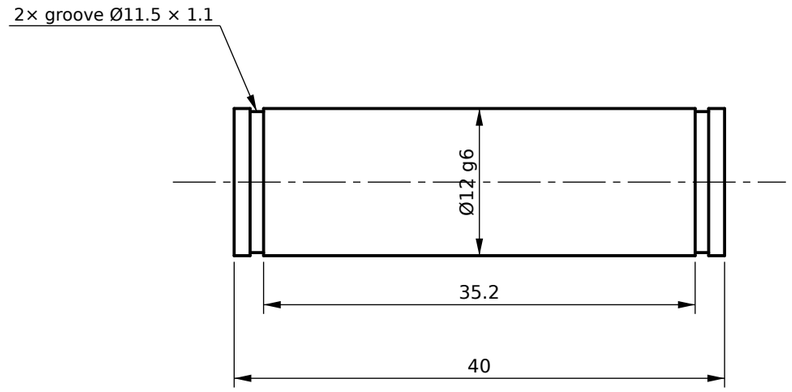


SIDE VIEW

NOTES

1. Screws into the handle hub MP-113 (M12 × 25 tapped) until the Ø20 shoulder seats; lock with Loctite 270.
2. Grip: red TPR sleeve Ø32 × 120 pushed on over the last 120 mm with rubber adhesive (bought).
3. Ends: 1 × 45° chamfer. Shoulder at the thread: square within 0.05.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 17 / 30
TITLE: **HANDLE BAR**			
DRAWING NO.: **MP-115**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: C45 Ø20 bright bar (or 304 stainless)			
HEAT TREATMENT / FINISH: Black oxide (none if stainless)			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:

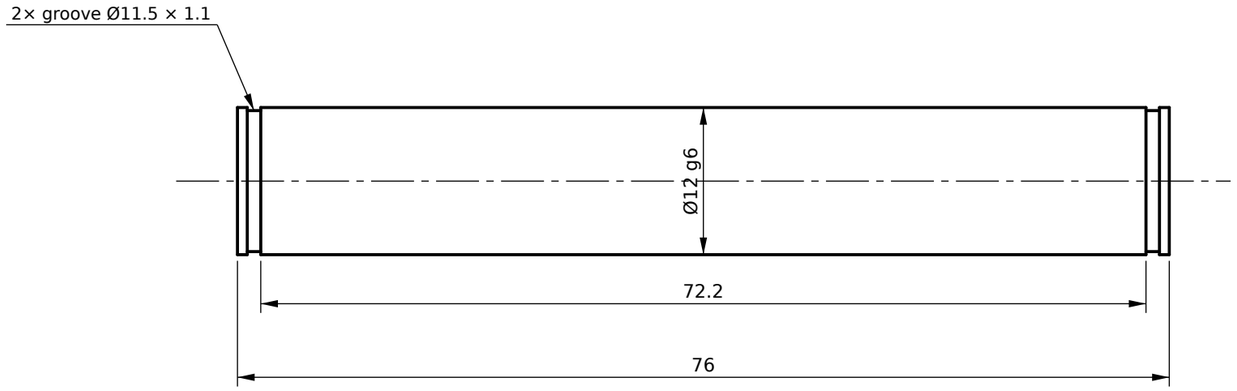


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 12: groove Ø11.5 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 17.6 from the pin centre (bearing faces at ±17.5).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 18 / 30
TITLE: **PIN A / PIN B (TOGGLE)**			
DRAWING NO.: **MP-116**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) Ø14 bar, or hardened ground dowel stock Ø12 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø12 g6, Ra 0.4; grease on assembly			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

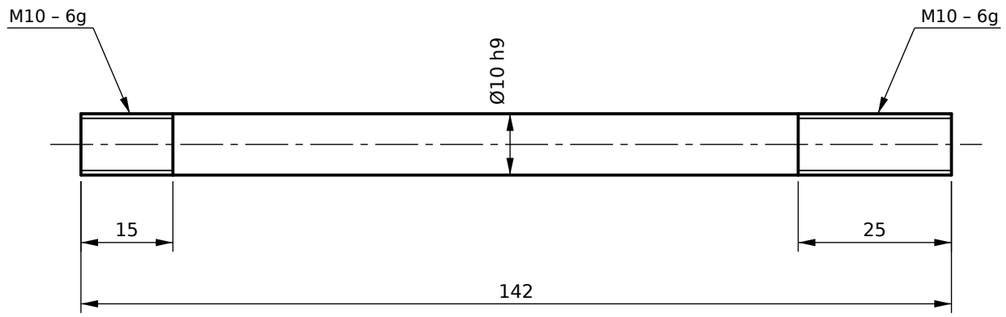


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 12: groove Ø11.5 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 36.1 from the pin centre (bearing faces at ±36).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 19 / 30
TITLE: **PIN O (HANDLE PIVOT)**			
DRAWING NO.: **MP-117**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) Ø14 bar, or hardened ground dowel stock Ø12 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø12 g6, Ra 0.4; grease on assembly			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

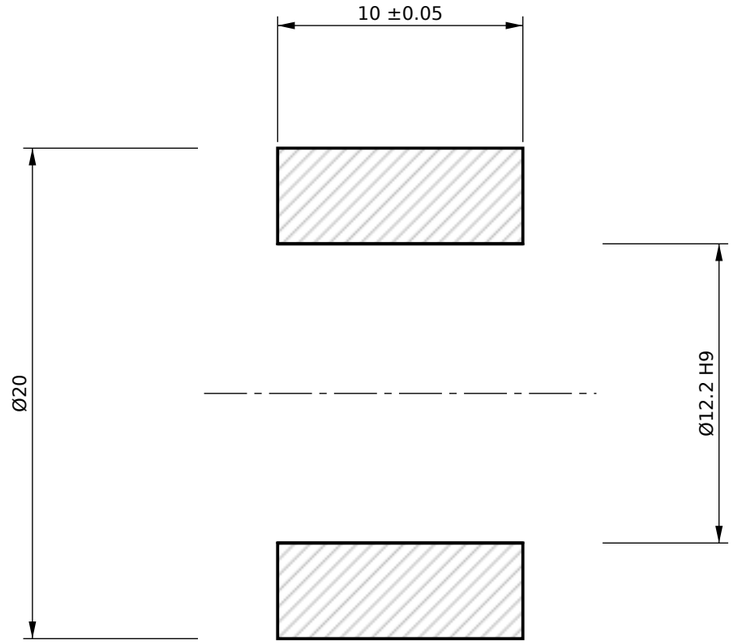


SIDE VIEW

NOTES

1. Bottom thread screws into the ram MP-109 (M10 × 15) with Loctite 243.
2. Top: return spring (music wire Ø2.5, OD 22, free 90), Ø30 washer and M10 nylock nut; set spring length 80 at handle-up.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 20 / 30
TITLE: **RETURN ROD**			
DRAWING NO.: **MP-118**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: C45 Ø12 bright bar			
HEAT TREATMENT / FINISH: Zinc plate 8 µm			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



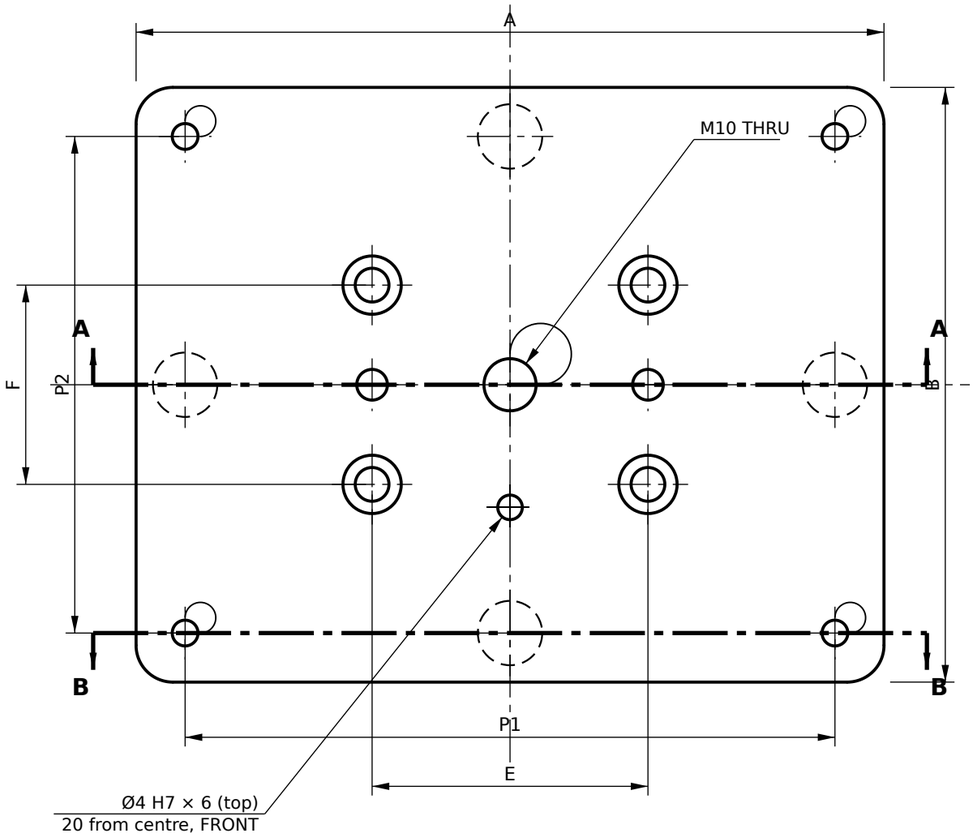
SECTION

NOTES

1. Fits on pin O between the handle hub and each pivot cheek. Faces parallel within 0.02.
2. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
3. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 21 / 30
TITLE: **PIVOT SPACER**			
DRAWING NO.: **MP-119**	QTY: 2	SCALE: 4:1	UNITS: mm
MATERIAL: CuSn8 bronze tube Ø22 × Ø10, or bar			
HEAT TREATMENT / FINISH: None			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A



SECTION B–B (Z = ±P2/2)

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	P1	P2	E	F
5050	50	50	82	82	66	66	25	25
6363	63.5	63.5	95.5	95.5	79.5	79.5	31.75	31.75
5380	80	53	112	85	96	69	40	26.5
6590	90	65	122	97	106	81	45	32.5

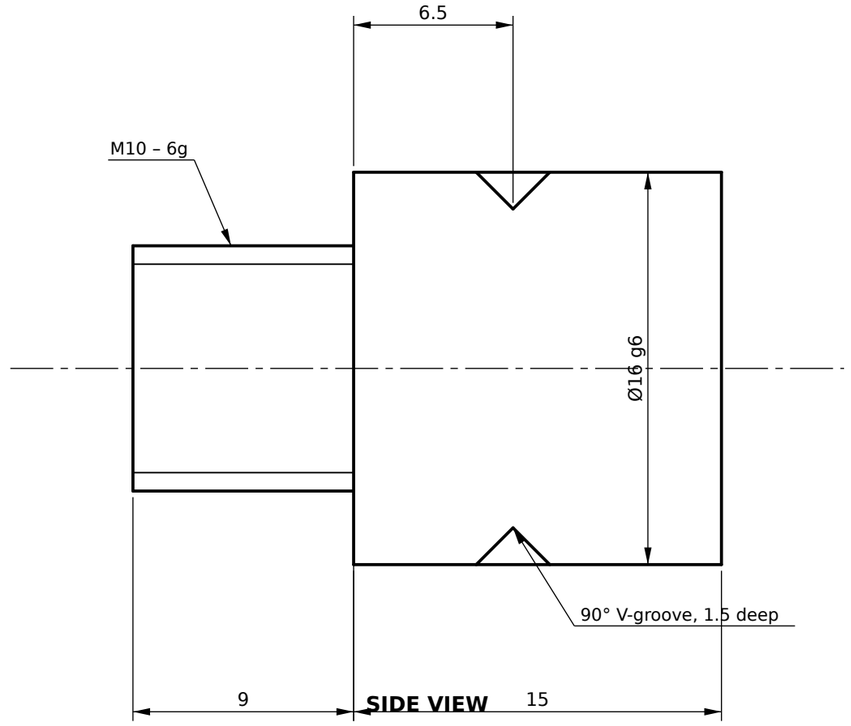
Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 32 (outside length)
B = Z + 32 (outside width)
P1 = X + 16 (pitch, retention screws / spring pockets in X)
P2 = Z + 16 (pitch in Z)
E = X / 2 (pitch, pedestal/plug screws in X)
F = Z / 2 (pitch in Z)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Top face (die seat against holder/shims) and bottom face parallel 0.01, flat 0.01.
4. Centre: M10 THRU for shank MP-202. Plug: 4 × Ø5.5 THRU, CBORE Ø9.5 × 5.5 from top (M5 × 20 ISO 4762) + 2 × Ø5 H7 THRU dowels (Ø5 × 20 m6).
5. Frame retention: 4 × M5 THRU (shoulder screws ISO 7379-6 × 10). Spring pockets 4 × Ø10.5 × 5 deep from the BOTTOM.
6. Orientation pin hole Ø4 H7 × 6 deep in the top face, 20 from centre toward the FRONT; press in Ø4 m6 × 10.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **A**	SHEET: 22 / 30	
TITLE: **UPPER DIE BODY**			
DRAWING NO.: **MP-201**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: 1.2312 (P20+S) prehardened 28–32 HRC, or C45 QT			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



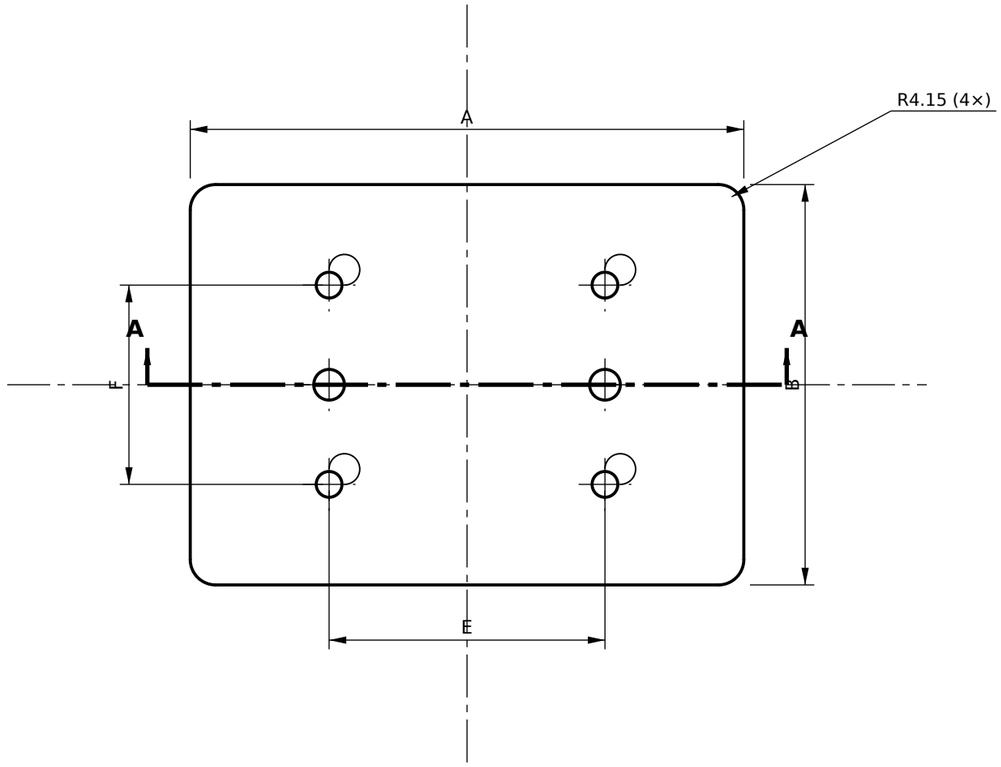
SIDE VIEW

NOTES

1. Screw into the die body MP-201 (M10 THRU) with Loctite 243 until the Ø16 face seats; face square to axis 0.01.
2. V-groove takes the cone-point lock screw in the die holder MP-111 (groove centre 6.5 from the seating face).
3. Top end chamfer 1 × 45° (lead-in).
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 23 / 30
TITLE: **UPPER DIE SHANK**			
DRAWING NO.: **MP-202**	QTY: 1 per die set (4)	SCALE: 4:1	UNITS: mm
MATERIAL: 42CrMo4 Ø20 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

SIZE	X	Z	A	B	E	F
5050	50	50	50.3	50.3	25	25
6363	63.5	63.5	63.8	63.8	31.75	31.75
5380	80	53	80.3	53.3	40	26.5
6590	90	65	90.3	65.3	45	32.5

Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 0.30 (plug length, ±0.01)

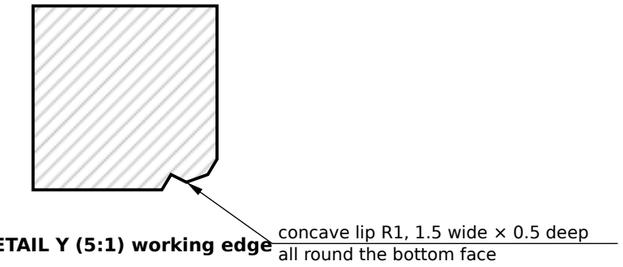
B = Z + 0.30 (plug width, ±0.01)

E = X / 2 (pitch, pedestal/plug screws in X)

F = Z / 2 (pitch in Z)



SECTION A–A



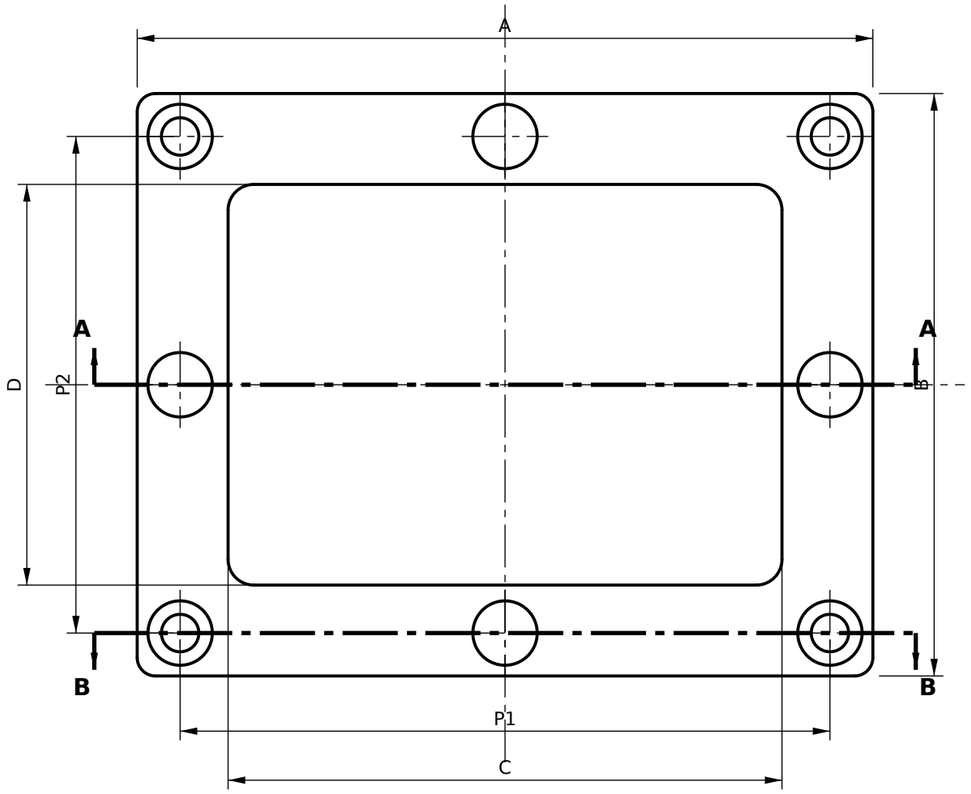
DETAIL Y (5:1) working edge

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Profile A × B with R4.15 corners: wire-EDM after hardening, ±0.01. Sliding fit in the upper frame opening (frame = plug + 0.05).
4. Top face: 4 × M5 × 10, 2 × Ø5 H7 × 8 (dowels into the body). Bottom (working) face flat 0.01, Ra 0.4, with lip per Detail Y.
5. Height 14.5 ±0.01: sets the 5.5 mm recess between plug face and frame face.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

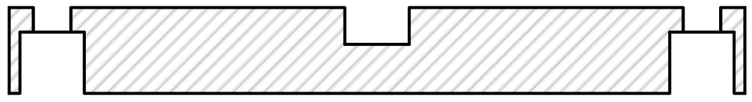
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 24 / 30
TITLE: **UPPER DIE PLUG**			
DRAWING NO.: **MP-203**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A



SECTION B–B (Z = ±P2/2)

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	C	D	P1	P2
5050	50	50	80	80	50.35	50.35	66	66
6363	63.5	63.5	93.5	93.5	63.85	63.85	79.5	79.5
5380	80	53	110	83	80.35	53.35	96	69
6590	90	65	120	95	90.35	65.35	106	81

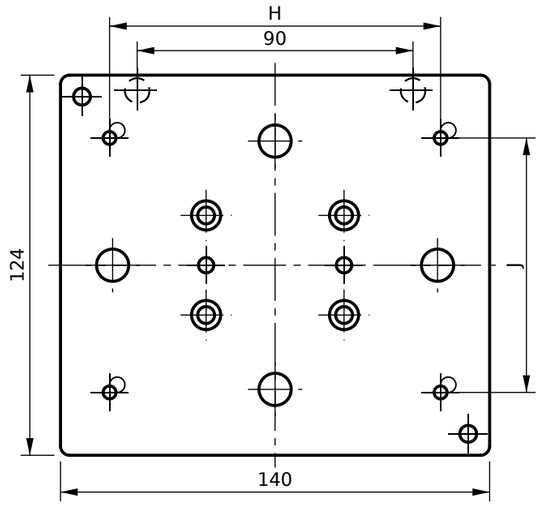
Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 30 (outside length)
B = Z + 30 (outside width)
C = X + 0.35 (opening, +0.01/0)
D = Z + 0.35 (opening, +0.01/0)
P1 = X + 16 (pitch, retention screws / spring pockets in X)
P2 = Z + 16 (pitch in Z)

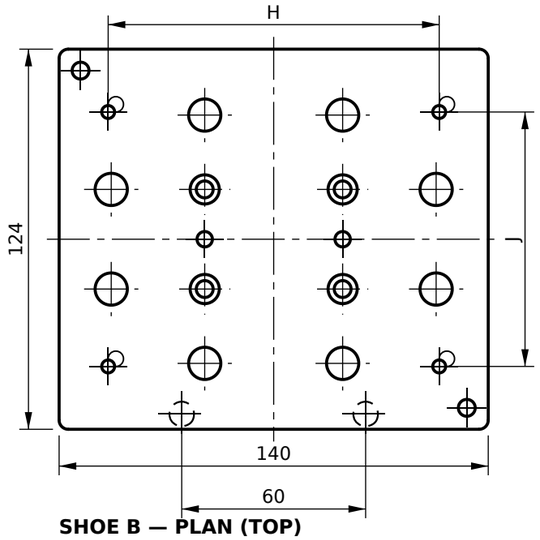
NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Opening C × D, R4.2 corners, THRU: wire-EDM after hardening, then lap to Ra 0.2. It must grip a wrapped shell by friction;
4. final-fit on the first prototype. Bottom inner edge R0.3 (do not leave sharp: it would cut the film).
5. Retention: 4 × Ø6.1 THRU, CBORE Ø10.5 × 10 from the BOTTOM (shoulder screw heads sit 5.5 inside the face).
6. Spring pockets 4 × Ø10.5 × 6 deep from the TOP. Faces parallel 0.01.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 25 / 30
TITLE: **UPPER DIE FRAME**			
DRAWING NO.: **MP-204**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED



SHOE A — PLAN (TOP)



SHOE B — PLAN (TOP)

Locating dowels: A at (±45, −57) = back; B at (±30, +57) = front.

DIMENSION TABLE (mm)

SIZE	X	Z	H	J	P1	P2	E	F
5050	50	50	68	68	66	66	25	25
6363	63.5	63.5	81.5	81.5	79.5	79.5	31.75	31.75
5380	80	53	98	71	96	69	40	26.5
6590	90	65	108	83	106	81	45	32.5

H = X + 18, J = Z + 18 (shoulder-screw pitch)
P1 = X + 16, P2 = Z + 16 (spring-pocket pitch)
E = X / 2, F = Z / 2 (pedestal-screw pitch)
Size code = magnet face: 5050, 6363, 5380 (53 × 80), 6590 (65 × 90).



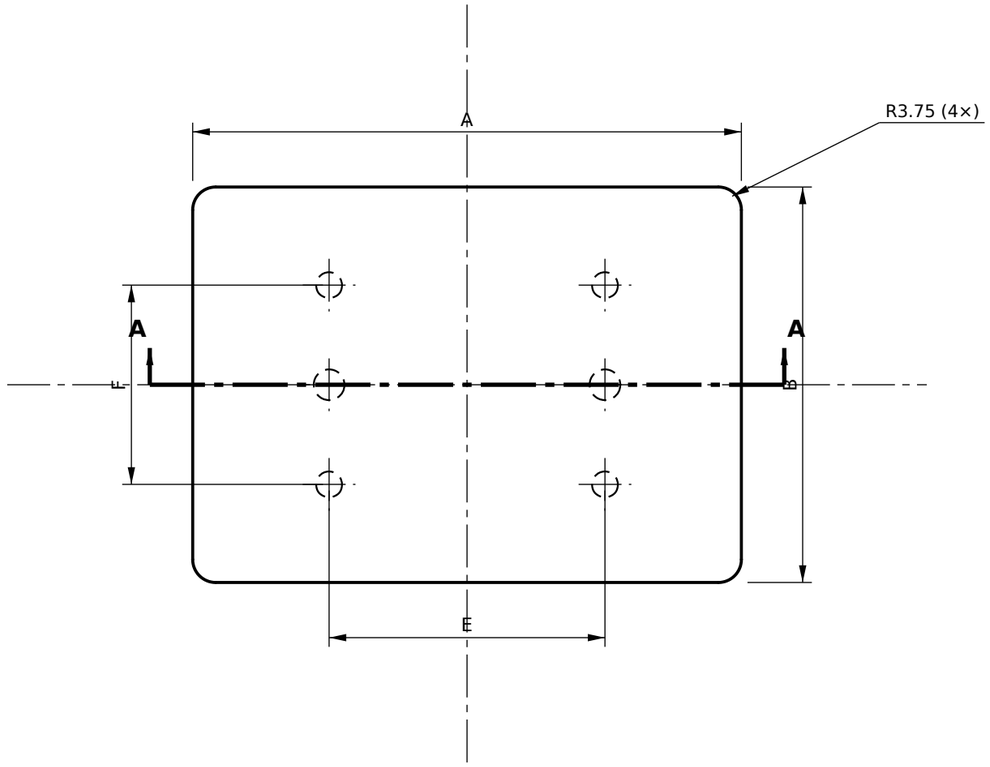
SECTION A–A, SHOE A, through Z = 0 (1:1)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. A and B differ only in the locating-dowel pattern and the spring pockets. Common: outline, thumb-screw holes, pedestal fixing.
4. Locating dowels: Ø8 H7 × 8 deep from the BOTTOM; press in Ø8 m6 × 16 (8 protrudes into the die table).
5. Shoulder screws (frame retention): 4 × M5 THRU at pitch H × J. Pedestal: 4 × Ø5.5 THRU, CBORE Ø9.5 × 5.5 BOTTOM + 2 × Ø5 H7 THRU.
6. Spring pockets Ø10.5 × 4 deep from the TOP: A = 4 at mid-sides (pitch P1, P2); B = 8 at (±P1/2, ±F/2) and (±E/2, ±P2/2).
7. Thumb-screw holes Ø5.5 THRU at (+63, +55) and (−63, −55) from the shoe centre (+Z = front). Faces parallel 0.01.
8. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
9. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 26 / 30
TITLE: **LOWER DIE SHOE A / B**			
DRAWING NO.: **MP-205**	QTY: 2 per set (A + B)	SCALE: 1:2 / 1:1	UNITS: mm
MATERIAL: C45 plate 12 mm (blank 145 × 130), faced to 10; or 6061-T6			
HEAT TREATMENT / FINISH: Black oxide; paint front edge band: A green, B red			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	E	F
5050	50	50	49.5	49.5	25	25
6363	63.5	63.5	63	63	31.75	31.75
5380	80	53	79.5	52.5	40	26.5
6590	90	65	89.5	64.5	45	32.5

Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X − 0.5 (±0.01)

B = Z − 0.5 (±0.01)

E = X / 2 (pitch, pedestal/plug screws in X)

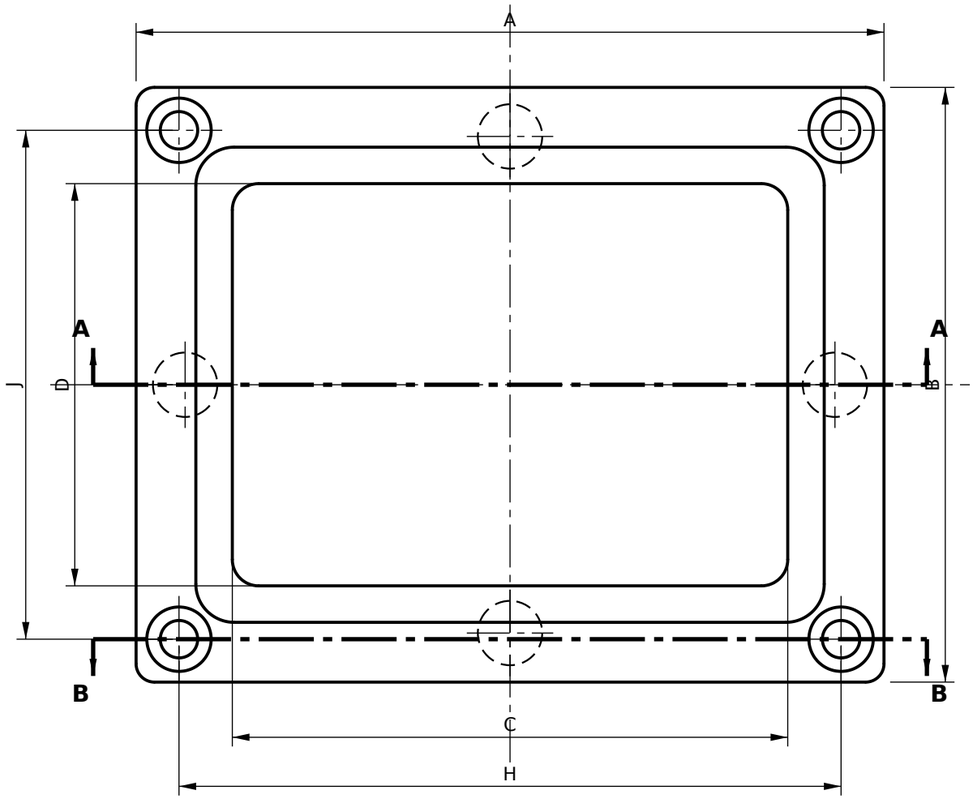
F = Z / 2 (pitch in Z)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. The tinplate shell sits face-up over this pedestal. Top face flat 0.01, Ra 0.4; top edge R0.3.
4. Bottom: 4 × M5 × 10, 2 × Ø5 H7 × 8. Height 21.8 ±0.02.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

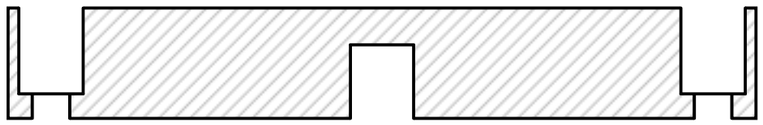
PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **A**	SHEET: 27 / 30	
TITLE: **PICKUP PEDESTAL (DIE A)**			
DRAWING NO.: **MP-206**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A



SECTION B–B (Z = ±J/2)

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	C	D	G	R	H	J	P1	P2
5050	50	50	82	82	50.6	50.6	62.5	62.5	68	68	66	66
6363	63.5	63.5	95.5	95.5	64.1	64.1	76	76	81.5	81.5	79.5	79.5
5380	80	53	112	85	80.6	53.6	92.5	65.5	98	71	96	69
6590	90	65	122	97	90.6	65.6	102.5	77.5	108	83	106	81

Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

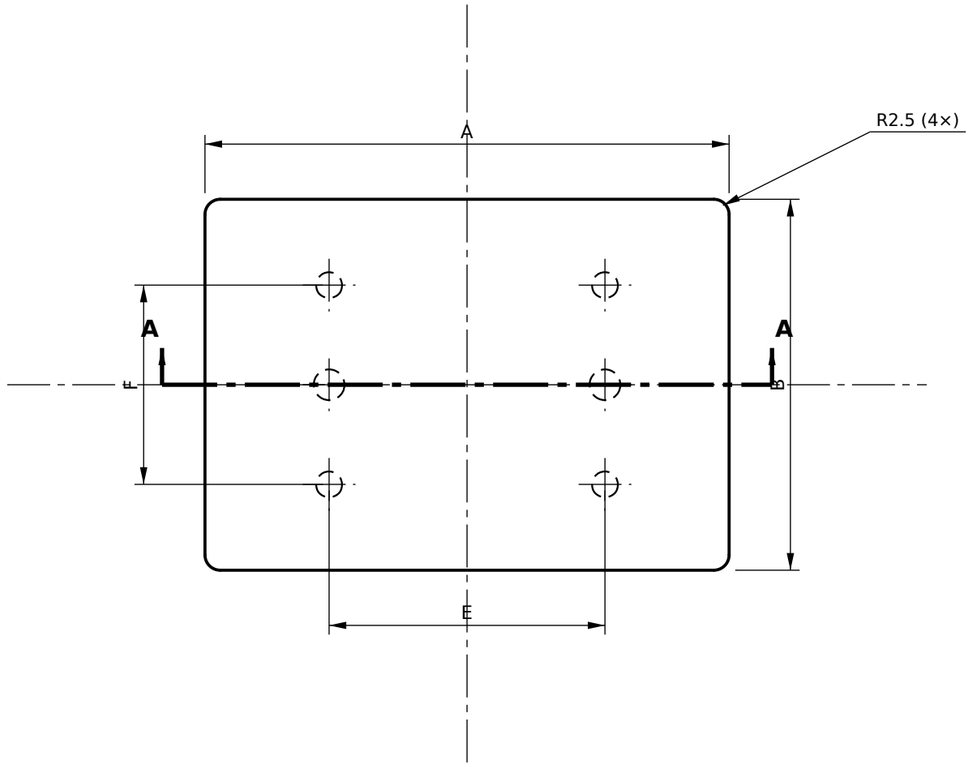
A = X + 32 (outside length)
B = Z + 32 (outside width)
C = X + 0.6 (opening)
D = Z + 0.6 (opening)
G = X + 12.5 (film recess, 0.6 deep)
R = Z + 12.5 (film recess)
H = X + 18 (pitch, shoulder screws in X)
J = Z + 18 (pitch, shoulder screws in Z)
P1 = X + 16 (pitch, spring pockets in X)
P2 = Z + 16 (pitch, spring pockets in Z)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Opening C × D, R4.3, THRU, wire-EDM; top inner edge R0.5. Film recess G × R, R6.25 corners, 0.6 deep, Ra 0.8.
4. Retention: 4 × Ø6.1 THRU, CBORE Ø10.5 × 14 from the TOP (shoulder screws ISO 7379-6 × 12 into the shoe).
5. Spring pockets 4 × Ø10.5 × 12 deep from the BOTTOM (mid-sides). Frame travel 8: it bottoms on the shoe.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 28 / 30
TITLE: **PICKUP FRAME (DIE A)**			
DRAWING NO.: **MP-207**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	E	F
5050	50	50	45.5	45.5	25	25
6363	63.5	63.5	59	59	31.75	31.75
5380	80	53	75.5	48.5	40	26.5
6590	90	65	85.5	60.5	45	32.5

Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X − 4.5 (±0.02)

B = Z − 4.5 (±0.02)

E = X / 2 (pitch, pedestal/plug screws in X)

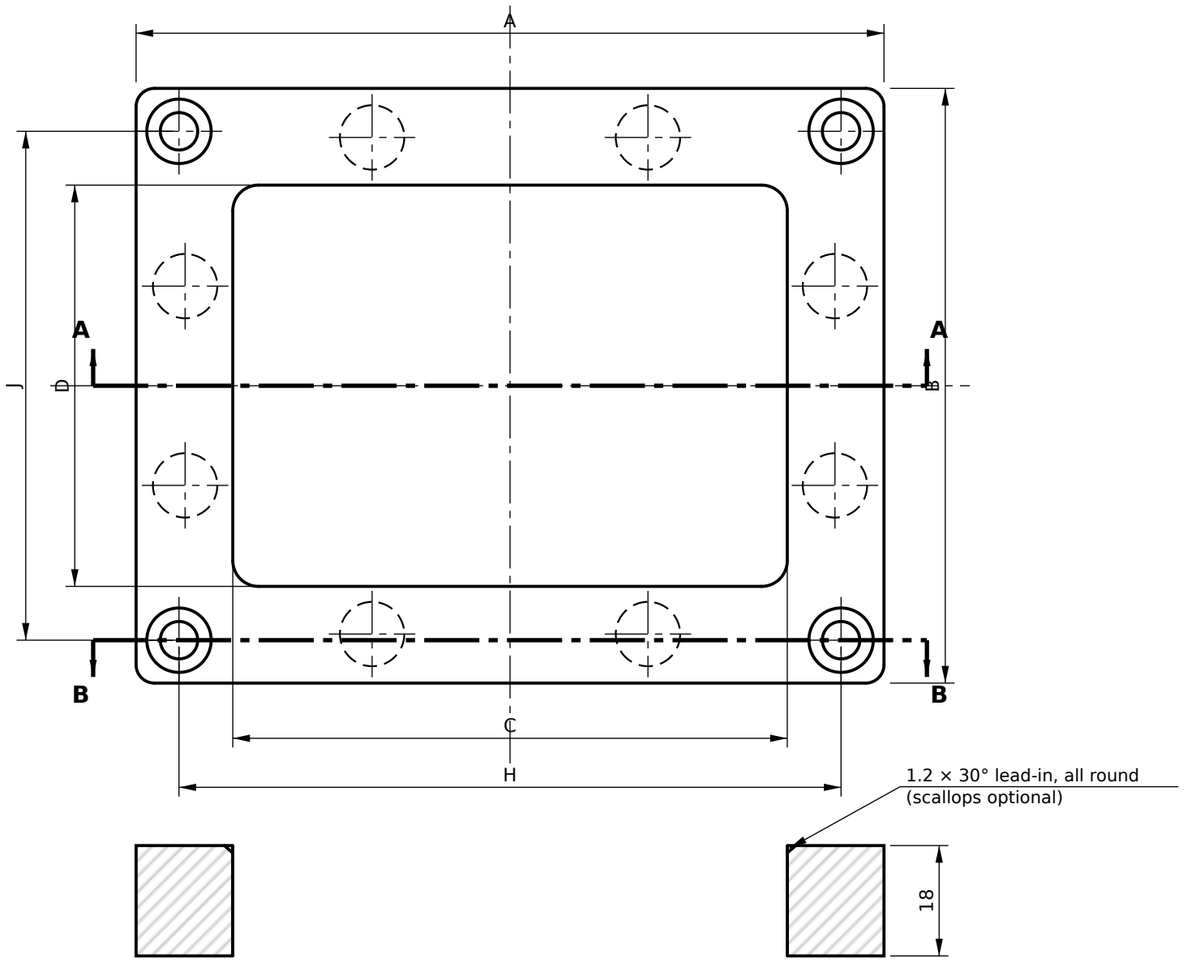
F = Z / 2 (pitch in Z)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Smaller than the back plate (X − 1.4) so the shell edge can roll under the back. Top face flat 0.01, Ra 0.4; top edge R0.5.
4. Bottom: 4 × M5 × 10, 2 × Ø5 H7 × 8. Height 17.0 ±0.02.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 29 / 30
TITLE: **CRIMP PEDESTAL (DIE B)**			
DRAWING NO.: **MP-208**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN:	CHECKED:

PLAN (TOP) — front at bottom



SECTION A–A



SECTION B–B (Z = ±J/2)

DIMENSION TABLE (mm)

SIZE	X	Z	A	B	C	D	H	J	P1	P2	E	F
5050	50	50	82	82	50.45	50.45	68	68	66	66	25	25
6363	63.5	63.5	95.5	95.5	63.95	63.95	81.5	81.5	79.5	79.5	31.75	31.75
5380	80	53	112	85	80.45	53.45	98	71	96	69	40	26.5
6590	90	65	122	97	90.45	65.45	108	83	106	81	45	32.5

Size code = magnet face size: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 32 (outside length)
B = Z + 32 (outside width)
C = X + 0.45 (opening, +0.01/0)
D = Z + 0.45 (opening, +0.01/0)
H = X + 18 (pitch, shoulder screws in X)
J = Z + 18 (pitch, shoulder screws in Z)
P1 = X + 16 (pitch, spring pockets in X)
P2 = Z + 16 (pitch, spring pockets in Z)
E = X / 2 (pitch, pedestal/plug screws in X)
F = Z / 2 (pitch in Z)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Opening C × D, R4.2, THRU, wire-EDM, lap Ra 0.2. Top lead-in 1.2 × 30° all round: folds the film under the shell edge.
4. Retention as MP-207. Spring pockets 8 × Ø10.5 × 12 deep from the BOTTOM at (±P1/2, ±F/2) and (±E/2, ±P2/2).
5. Crimp springs: total preload per the build pack (≈ 1.4–2.1 kN by size); cavity 24 at rest, 16 at full travel.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **A**	SHEET: 30 / 30
TITLE: **CRIMP FRAME (DIE B)**			
DRAWING NO.: **MP-209**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: Cr12MoV / 1.2379 (D2), or 7075-T6 for prototype dies			
HEAT TREATMENT / FINISH: Vacuum harden 58–60 HRC, temper ×2; wire-EDM profiles after hardening; working faces Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-26	DRAWN / CHECKED